

HW04 AI Audit Report

- Student ID: 23127027

I use AI tools for the following tasks: repository safety review, audit-workflow design, test planning, automation generation, human review support, debugging, and documentation.

Interaction 0001

AI tool

OpenAI Codex CLI 0.147.0 (model: gpt-5.6-sol)

ISO date/time
2026-08-17T09:11:28Z

Exact prompt

```text

Act as a documentation assistant for HW04 of student 23127027. Inspect only the existing ai-audit directory. Propose one concise Markdown audit-entry template with exactly five fields: AI tool, ISO date/time, Exact prompt, Exact AI output, Human review/action. Do not modify files and do not invent evidence.

## Exact AI output

```text

```markdown

## AI tool

[Record the tool used]

## ISO date/time  
[YYYY-MM-DDThh:mm:ssZ]

## Exact prompt  
[Paste verbatim]

## Exact AI output  
[Paste verbatim]

## Human review/action  
[Record verification, edits, acceptance, or rejection]

No files were modified.

## Human review/action

Em đã đối chiếu ba file, xác nhận prompt và output được lưu nguyên văn, template có đủ năm trường, không có evidence bị bịa và không có source file nào bị sửa.

# Interaction 0002

## AI tool

OpenAI Codex CLI 0.147.0

## ISO date/time  
2026-08-18T01:27:52Z

## Exact prompt

````text

Act as an implementation-inventory assistant for HW04 student 23127027. Inspect only repository documentation and source files relevant to FR-01 Account registration, FR-07 Shopping cart, and FR-12 Access control. Produce a concise Markdown table with columns: Feature, frontend files/components, backend routes or middleware, client state or data storage, and evidence as file:line. Do not propose test cases. Do not modify files. Do not label any discrepancy as a bug. Explicitly mark anything not found.

Exact AI output

````text

Implementation inventory based only on scoped documentation and source files. "Not found" means no corresponding implementation was located in those files.

| Feature                    | Frontend files/components                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Backend routes or middleware                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Client state or data storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Evidence as file:line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01 Account registration | Web: `Register` form and `./register` route; collects name, email, and password, then redirects to login. Mobile: `renderRegister` and `handleRegister` with local password validation. **Not found:** confirmation-password field in either client; admin registration UI.   `POST /api/register` inserts name, email, and password into `users`. **Not found:** route middleware, backend input/password-strength validation, password hashing, or explicit email-uniqueness enforcement.   Web and mobile form values use transient React state. Registered accounts are stored in SQLite's `users` table; its `role` defaults to `user`. Registration-state persistence: **Not found**.   Requirements: `README.md:30-36`.                                                                                                                                                   | Web: `./register` route; collects name, email, and password, then redirects to login. Mobile: `renderRegister` and `handleRegister` with local password validation. **Not found:** confirmation-password field in either client; admin registration UI.   `POST /api/register` inserts name, email, and password into `users`. **Not found:** route middleware, backend input/password-strength validation, password hashing, or explicit email-uniqueness enforcement.   Web and mobile form values use transient React state. Registered accounts are stored in SQLite's `users` table; its `role` defaults to `user`. Registration-state persistence: **Not found**.   Requirements: `README.md:30-36`.                                                                                                                                                                       | Web: `./register` route; collects name, email, and password, then redirects to login. Mobile: `renderRegister` and `handleRegister` with local password validation. **Not found:** confirmation-password field in either client; admin registration UI.   `POST /api/register` inserts name, email, and password into `users`. **Not found:** route middleware, backend input/password-strength validation, password hashing, or explicit email-uniqueness enforcement.   Web and mobile form values use transient React state. Registered accounts are stored in SQLite's `users` table; its `role` defaults to `user`. Registration-state persistence: **Not found**.   Requirements: `README.md:30-36`.                                                                                                                                                                       | Web: `./register` route; collects name, email, and password, then redirects to login. Mobile: `renderRegister` and `handleRegister` with local password validation. **Not found:** confirmation-password field in either client; admin registration UI.   `POST /api/register` inserts name, email, and password into `users`. **Not found:** route middleware, backend input/password-strength validation, password hashing, or explicit email-uniqueness enforcement.   Web and mobile form values use transient React state. Registered accounts are stored in SQLite's `users` table; its `role` defaults to `user`. Registration-state persistence: **Not found**.   Requirements: `README.md:30-36`.                                                                                                                                                                       | Web: `./register` route; collects name, email, and password, then redirects to login. Mobile: `renderRegister` and `handleRegister` with local password validation. **Not found:** confirmation-password field in either client; admin registration UI.   `POST /api/register` inserts name, email, and password into `users`. **Not found:** route middleware, backend input/password-strength validation, password hashing, or explicit email-uniqueness enforcement.   Web and mobile form values use transient React state. Registered accounts are stored in SQLite's `users` table; its `role` defaults to `user`. Registration-state persistence: **Not found**.   Requirements: `README.md:30-36`.                                                                                                                                                                       |
| FR-07 Shopping cart        | Web: `Home` and `ProductDetail` add items; `Cart` displays/removes items and total; `CartProvider` supplies cart operations. Mobile: product detail, `addToCart`, and `renderCart`; duplicate product IDs are merged. **Not found:** deletion confirmation or empty-cart illustration in web/mobile; web quantity adjustment controls.   Authenticated `GET /api/cart` and `POST /api/cart`; carts are held in the server's `userCarts` object. **Not found:** update/delete cart routes or calls to these cart routes from the web/mobile clients.   Web: in-memory `CartContext` state; adding always appends. Mobile: in-memory component state with duplicate-item merging. Server: in-memory cart arrays keyed by user ID. Persistent cart table/storage: **Not found**.   Requirements: `README.md:93-100`; API docs: `api_specification.md:110-127`.                      | Web: `Home` and `ProductDetail` add items; `Cart` displays/removes items and total; `CartProvider` supplies cart operations. Mobile: product detail, `addToCart`, and `renderCart`; duplicate product IDs are merged. **Not found:** deletion confirmation or empty-cart illustration in web/mobile; web quantity adjustment controls.   Authenticated `GET /api/cart` and `POST /api/cart`; carts are held in the server's `userCarts` object. **Not found:** update/delete cart routes or calls to these cart routes from the web/mobile clients.   Web: in-memory `CartContext` state; adding always appends. Mobile: in-memory component state with duplicate-item merging. Server: in-memory cart arrays keyed by user ID. Persistent cart table/storage: **Not found**.   Requirements: `README.md:93-100`; API docs: `api_specification.md:110-127`.                      | Web: `Home` and `ProductDetail` add items; `Cart` displays/removes items and total; `CartProvider` supplies cart operations. Mobile: product detail, `addToCart`, and `renderCart`; duplicate product IDs are merged. **Not found:** deletion confirmation or empty-cart illustration in web/mobile; web quantity adjustment controls.   Authenticated `GET /api/cart` and `POST /api/cart`; carts are held in the server's `userCarts` object. **Not found:** update/delete cart routes or calls to these cart routes from the web/mobile clients.   Web: in-memory `CartContext` state; adding always appends. Mobile: in-memory component state with duplicate-item merging. Server: in-memory cart arrays keyed by user ID. Persistent cart table/storage: **Not found**.   Requirements: `README.md:93-100`; API docs: `api_specification.md:110-127`.                      | Web: `Home` and `ProductDetail` add items; `Cart` displays/removes items and total; `CartProvider` supplies cart operations. Mobile: product detail, `addToCart`, and `renderCart`; duplicate product IDs are merged. **Not found:** deletion confirmation or empty-cart illustration in web/mobile; web quantity adjustment controls.   Authenticated `GET /api/cart` and `POST /api/cart`; carts are held in the server's `userCarts` object. **Not found:** update/delete cart routes or calls to these cart routes from the web/mobile clients.   Web: in-memory `CartContext` state; adding always appends. Mobile: in-memory component state with duplicate-item merging. Server: in-memory cart arrays keyed by user ID. Persistent cart table/storage: **Not found**.   Requirements: `README.md:93-100`; API docs: `api_specification.md:110-127`.                      | Web: `Home` and `ProductDetail` add items; `Cart` displays/removes items and total; `CartProvider` supplies cart operations. Mobile: product detail, `addToCart`, and `renderCart`; duplicate product IDs are merged. **Not found:** deletion confirmation or empty-cart illustration in web/mobile; web quantity adjustment controls.   Authenticated `GET /api/cart` and `POST /api/cart`; carts are held in the server's `userCarts` object. **Not found:** update/delete cart routes or calls to these cart routes from the web/mobile clients.   Web: in-memory `CartContext` state; adding always appends. Mobile: in-memory component state with duplicate-item merging. Server: in-memory cart arrays keyed by user ID. Persistent cart table/storage: **Not found**.   Requirements: `README.md:93-100`; API docs: `api_specification.md:110-127`.                      |
| FR-12 Access control       | Separate admin `App` provides a login gate, checks `res.data.user.role === "admin"`, and renders admin functions only when a token exists. **Not found:** admin-access components in the customer web or mobile clients.   Login JWT contains `id` and `role`; `authenticateToken` verifies JWTs. Admin routes use this token middleware. Product mutations have no middleware; category and admin coupon mutations use token authentication. Role-checking backend middleware or a server-side `role === "admin"` authorization check: **Not found**.   Admin token is held in React state and persisted as `localStorage.adminToken`; it becomes Axios's default Authorization header. Server roles reside in SQLite and are embedded in JWTs. Persisted client-side role state: **Not found**.   Requirements: `README.md:174-179`; API docs: `api_specification.md:171-181`. | Separate admin `App` provides a login gate, checks `res.data.user.role === "admin"`, and renders admin functions only when a token exists. **Not found:** admin-access components in the customer web or mobile clients.   Login JWT contains `id` and `role`; `authenticateToken` verifies JWTs. Admin routes use this token middleware. Product mutations have no middleware; category and admin coupon mutations use token authentication. Role-checking backend middleware or a server-side `role === "admin"` authorization check: **Not found**.   Admin token is held in React state and persisted as `localStorage.adminToken`; it becomes Axios's default Authorization header. Server roles reside in SQLite and are embedded in JWTs. Persisted client-side role state: **Not found**.   Requirements: `README.md:174-179`; API docs: `api_specification.md:171-181`. | Separate admin `App` provides a login gate, checks `res.data.user.role === "admin"`, and renders admin functions only when a token exists. **Not found:** admin-access components in the customer web or mobile clients.   Login JWT contains `id` and `role`; `authenticateToken` verifies JWTs. Admin routes use this token middleware. Product mutations have no middleware; category and admin coupon mutations use token authentication. Role-checking backend middleware or a server-side `role === "admin"` authorization check: **Not found**.   Admin token is held in React state and persisted as `localStorage.adminToken`; it becomes Axios's default Authorization header. Server roles reside in SQLite and are embedded in JWTs. Persisted client-side role state: **Not found**.   Requirements: `README.md:174-179`; API docs: `api_specification.md:171-181`. | Separate admin `App` provides a login gate, checks `res.data.user.role === "admin"`, and renders admin functions only when a token exists. **Not found:** admin-access components in the customer web or mobile clients.   Login JWT contains `id` and `role`; `authenticateToken` verifies JWTs. Admin routes use this token middleware. Product mutations have no middleware; category and admin coupon mutations use token authentication. Role-checking backend middleware or a server-side `role === "admin"` authorization check: **Not found**.   Admin token is held in React state and persisted as `localStorage.adminToken`; it becomes Axios's default Authorization header. Server roles reside in SQLite and are embedded in JWTs. Persisted client-side role state: **Not found**.   Requirements: `README.md:174-179`; API docs: `api_specification.md:171-181`. | Separate admin `App` provides a login gate, checks `res.data.user.role === "admin"`, and renders admin functions only when a token exists. **Not found:** admin-access components in the customer web or mobile clients.   Login JWT contains `id` and `role`; `authenticateToken` verifies JWTs. Admin routes use this token middleware. Product mutations have no middleware; category and admin coupon mutations use token authentication. Role-checking backend middleware or a server-side `role === "admin"` authorization check: **Not found**.   Admin token is held in React state and persisted as `localStorage.adminToken`; it becomes Axios's default Authorization header. Server roles reside in SQLite and are embedded in JWTs. Persisted client-side role state: **Not found**.   Requirements: `README.md:174-179`; API docs: `api_specification.md:171-181`. |

```
>Admin client: `frontend-admin/src/App.jsx:7-10`, `frontend-admin/src/App.jsx:34-70`, `frontend-admin/src/App.jsx:188-215`, `frontend-admin/src/App.jsx:263-268`
.
>JWT/middleware: `backend/server.js:46-52`, `backend/server.js:100-110`.
Product routes: `backend/server.js:167-196`; category routes: `backend/server.js:249-278`; admin routes: `backend/server.js:199-241`, `backend/server.js:457-488`,
`backend/server.js:494-568`.
>Role storage: `backend/database.js:48-61`. |
```

## Human review/action

Em đã đối chiếu các dẫn chiếu web và backend cho FR-01, FR-07 và FR-12; các nhận định chính phù hợp source. Em loại phần frontend-mobile khỏi test basis HW04 vì phạm vi Playwright là web. Các discrepancy được giữ làm test target và chưa được báo cáo là defect khi chưa có execution evidence.

# Interaction 0003

## AI tool

OpenAI Codex CLI 0.147.0

## ISO date/time  
2026-08-18T01:45:03Z

## Exact prompt

```text

Act as a test-design assistant for HW04 student 23127027. Create exactly one file: docs/hw04-traceability-test-matrix.md. Do not modify any other file. Base the matrix on README.md version 2.0 requirements for FR-01, FR-07, FR-12, relevant FR-21 through FR-24, and SEC-02, SEC-03, SEC-06. Use api_specification.md only as supporting API contract. Exclude frontend-mobile and all HW02 test cases/results. Create exactly 36 planned automated test cases: exactly 12 IDs FR01-TC01 through FR01-TC12, exactly 12 IDs FR07-TC01 through FR07-TC12, and exactly 12 IDs FR12-TC01 through FR12-TC12. For each case include: source requirement, UI or API layer, test-design technique or data partition, objective and external-data intent, expected result derived from the requirement, assertion pattern, Chromium/Firefox/Safari/WebKit applicability, and mutation/cleanup note. Plan separate JSON data files tests/data/fr01-registration.json, tests/data/fr07-cart.json, and tests/data/fr12-access-control.json; never propose inline test-case arrays or objects in spec files. Ensure at least three distinct assertion patterns within each feature. Mark implementation discrepancies only as test targets, never as confirmed defects. Mark unsupported expectations as EVIDENCE MISSING instead of inventing them. For FR-12 distinguish authentication from authorization, include ordinary-user and admin-token coverage, and mark data-mutating checks for controlled backup/restore or cleanup. Add concise sections for evidence hierarchy, execution policy, and requirement coverage. Do not claim tests were executed or passed. Do not run servers, tests, package installation, Git commands, or commits. If the target file already exists, do not overwrite it and report that fact.

Exact AI output

```text

Created [docs/hw04-traceability-test-matrix.md](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/docs/hw04-traceability-test-matrix.md).

It contains exactly 36 planned cases:

- 12 × FR01-TC01-FR01-TC12
- 12 × FR07-TC01-FR07-TC12
- 12 × FR12-TC01-FR12-TC12

No other files were modified, and no tests, servers, installations, or Git commands were run.

## Human review/action

Em đã kiểm tra đủ 36 case và không chấp nhận draft 0003 ở trạng thái hiện tại. Em yêu cầu loại suy diễn ảnh sản phẩm khỏi FR07-TC01; đổi FR07-TC03 thành kiểm tra gộp sản phẩm hoàn toàn qua UI; làm oracle badge FR07-TC12 xác định bằng trạng thái ban đầu không có sản phẩm và thêm đúng một đơn vị; bổ sung chính sách 9 feature-browser runs, 9 HTML reports với student ID và ISO timestamp từ execution timer; đồng thời cập nhật coverage FR-12 gồm TC11-TC12. Không thay đổi số lượng 12 case cho mỗi feature và chưa gọi discrepancy nào là defect.

# Interaction 0004

## AI tool

OpenAI Codex CLI 0.147.0

## ISO date/time

2026-08-18T01:58:09Z

## Exact prompt

```text

Act as a correction-only editor for docs/hw04-traceability-test-matrix.md. Modify exactly that one file and no other file. Preserve the existing 36-case structure, IDs, evidence hierarchy, and all content not required by these corrections. Apply only these human-reviewed corrections: (1) In FR07-TC01 remove the unsupported requirement that Cart must display product images and remove its FR-24 image-alt mapping; keep the SRS-required cart columns, h1, and currency checks. (2) Change FR07-TC03 to a UI-only same-product test using external data: start with the target product absent, add the same product twice through visible web UI, then assert exactly one cart row for that product, quantity 2, and consistent line/cart totals. Do not use or cross-check backend /api/cart in this case. (3) Make FR07-TC12 deterministic: start with the target product absent and a known initial badge, add exactly one unit through UI, assert visual feedback and badge increment exactly 1; because this case uses a product page/card image, assert any displayed product image has a non-empty descriptive alt. (4) Add execution policy requiring exactly 9 independent feature-browser runs: FR-01, FR-07, FR-12 each on Chromium, Firefox, WebKit; each run must generate its own genuine HTML report displaying exactly Run by: 23127027 and an ISO timestamp generated during execution, with no post-editing. (5) Update requirement coverage so FR-12 includes FR12-TC01 through FR12-TC12 and FR-24 no longer maps FR07-TC01 while retaining the corrected FR07-TC12 mapping. Keep exactly 12 IDs per feature, external JSON data, at least three assertion patterns per feature, C/F/W applicability, EVIDENCE MISSING handling, mutation safety, and no claimed execution or defects. Do not run tests, servers, installations, Git commands, or commits.

Exact AI output

```text

Updated only [docs/hw04-traceability-test-matrix.md](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/docs/hw04-traceability-test-matrix.md).

Applied all five requested corrections and verified the structure remains exactly 36 cases: 12 each for FR-01, FR-07, and FR-12. No tests, servers, installation, or Git commands were run.

## Human review/action

Em đã so sánh corrected matrix với rejected artifact 0003 và xác nhận Codex chỉ áp dụng các correction đã yêu cầu. Matrix giữ đúng 12 case cho mỗi feature, dùng external JSON, có ít nhất ba assertion patterns cho mỗi feature, áp dụng Chromium/Firefox/WebKit, quy định 9 run và 9 genuine HTML reports với Run by: 23127027 cùng ISO timestamp sinh trong execution. Không có source SUT bị sửa và chưa có pass/fail hoặc defect nào được tuyên bố.

# Interaction 0005

## AI tool

OpenAI Codex CLI 0.147.0

## ISO date/time  
2026-08-18T07:11:10Z

## Exact prompt

```
````text
Act as a Playwright infrastructure implementer for HW04 student 23127027. Work only at repository root. You may modify package.json and create exactly these files: playwright.config.ts, tsconfig.json, .env.example, tests/smoke/infrastructure.spec.ts. Do not modify .gitignore, package-lock.json, SUT source, documentation, audit files, or any backend/frontend package. Add package scripts for playwright test, test listing, infrastructure smoke, TypeScript noEmit checking, and opening the default HTML report. Configure exactly three Playwright projects named chromium, firefox, webkit with the matching desktop devices. Use testDir tests, strict deterministic defaults with fullyParallel false, workers 1, retries 0, forbidOnly on CI, and reasonable test/expect timeouts. Use PW_WEB_URL with default http://localhost:5173 as baseURL. Configure list plus HTML reporters. Read PW_REPORT_DIR with default playwright-report for the HTML output folder and never post-edit reports. At config evaluation, generate one new Date().toISOString() value and use it both in config metadata and the visible HTML reporter title. The title must contain the exact text Run by: 23127027 and the ISO timestamp. Metadata must include Run by with value 23127027, ISO timestamp, and PW_RUN_LABEL with a safe local default. Keep screenshots only on failure, traces retained on failure, and video off. The committed .env.example may contain only non-secret PW_WEB_URL, PW_ADMIN_URL, PW_API_URL, PW_REPORT_DIR, and PW_RUN_LABEL examples. Create one infrastructure smoke spec that does not contact the SUT: it must run in all three projects, verify the metadata values and ISO timestamp shape, set simple local page content, and use a web-first assertion to prove the selected browser can execute. Do not add feature tests or test-data arrays. Do not install packages, browsers, run tests, start servers, use Git, or commit. After editing, summarize exactly which allowed files changed.
```

Exact AI output

```
````text
Implemented the Playwright infrastructure without running tests, installing packages, starting servers, or using Git.
```

Changed exactly these allowed files:

- Modified [package.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/package.json)
- Created [playwright.config.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/playwright.config.ts)
- Created [tsconfig.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tsconfig.json)
- Created [.env.example](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/.env.example)

- Created [tests/smoke/infrastructure.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/smoke/infrastructure.spec.ts)

No prohibited files were edited.

## Human review/action

Em đã review package.json, Playwright config, TypeScript config, environment example và infrastructure smoke spec. Typecheck thành công, test listing có đúng ba browser projects và smoke execution thật pass 3/3. Em đã mở HTML report và xác nhận hiển thị Run by: 23127027, ISO timestamp 2026-08-18T07:20:21.847Z cùng kết quả Chromium, Firefox và WebKit. Report này chỉ dùng kiểm chứng infrastructure, không được tính vào chín feature-browser reports cuối và không có source SUT nào bị sửa.

# Interaction 0006

## AI tool

OpenAI Codex CLI 0.147.0

## ISO date/time

2026-08-18T12:20:40Z

## Exact prompt

```text

Implement only the first reviewed FR-01 Playwright increment for HW04 student 23127027. You may create exactly these files: tests/data/fr01-registration.json, tests/support/data-loader.ts, tests/pages/registration.page.ts, tests/features/fr01-registration.spec.ts. Do not modify any existing file, SUT source, configuration, package, documentation, or audit file. Use the authoritative requirements and docs/hw04-traceability-test-matrix.md. Implement exactly logical IDs FR01-TC01 through FR01-TC04; every test title must begin with its ID. Put all case-specific inputs, expected labels, omission variants, and email templates in the JSON file. The spec must not contain inline test-case arrays or objects. Load JSON from disk through a typed reusable loader. Generate collision-safe emails at runtime from external templates without committing live credentials. Create a registration page object with form-scoped Playwright locators and action methods; do not use fixed sleeps. TC01 must cover valid registration destination plus one h1, Vietnamese registration content, and blue positive submit behavior without inventing an exact shade. TC02 must cover required controls, visible asterisk markers, blank/one-field omission rejection, and application-error placement above submit when such an error is produced. TC03 must cover input type=email plus invalid and valid email-format partitions while avoiding assertions on browser-specific native validation wording. TC04 must cover the confirmation-password control, both password controls using type=password, exactly one h1, and top-to-bottom keyboard focus order. Missing required UI must cause clear failing assertions; do not conditionally bypass it. Do not use test.skip, test.fail, weak placeholder assertions, source-code assertions, or post-edit evidence. Do not run tests, servers, package installation, Git commands, or commits. After editing, summarize exactly the allowed files created and any design assumptions.

Exact AI output

```text

Implemented exactly FR01-TC01 through FR01-TC04 and created only:

- [fr01-registration.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr01-registration.json)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)

ort/data-loader.ts)

- [registration.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/test s/pages/registration.page.ts)
- [fr01-registration.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027 /tests/features/fr01-registration.spec.ts)

Design assumptions:

- Login destination is verified semantically through login-form content, since the exact redirect URL is unspecified.
- “Blue” means the computed color’s blue channel is dominant; no exact shade is asserted.
- Email partitions use native validity state, never browser-specific messages.
- Application-error placement is checked whenever an application error is rendered; native validation need not produce one.
- Focus traversal begins at the first form control and verifies visual and Tab order through submit.
- No documented safe account-deletion interface exists, so created accounts use collision-safe runtime emails from external templates.

Per instruction, I did not run tests, servers, package commands, Git commands, or post-edit verification.

## Human review/action

Em đã review code trước khi chạy và phát hiện locator getByLabel không phù hợp DOM hiện tại, exact button name có thể làm mất form locator, fill luôn phụ thuộc confirmation làm nhiều TC03, ID/title chưa nằm trong JSON và error locator có thể bỏ sót error thật. Em yêu cầu sửa automation để locator form-scoped theo cấu trúc label-parent-input, dùng accessible-name case-insensitive, cho phép điền đúng tập field cần kiểm tra, lấy ID/title từ external JSON và phát hiện error mà không sửa SUT hoặc né assertion. Chưa có test execution hoặc defect claim.

# Interaction 0007

## AI tool

OpenAI Codex CLI 0.147.0

## ISO date/time

2026-08-18T12:32:41Z

## Exact prompt

```text

Act as a correction-only editor for the rejected FR-01 increment. Modify exactly these four files and no others: tests/data/fr01-registration.json, tests/support/data-loader.ts, tests/pages/registration.page.ts, tests/features/fr01-registration.spec.ts. Preserve logical scope FR01-TC01 through FR01-TC04 and all hard requirement assertions. Apply these human-reviewed fixes only. Add id and title to each of the four top-level external JSON cases, type them, and construct every Playwright test title from those JSON values rather than hardcoding IDs/titles in the spec. Replace getByLabel-based registration and login input locators: source labels are unassociated siblings, so locate each label inside its form using an escaped anchored case-insensitive pattern, move to its containing element, and then locate the corresponding input. Use anchored case-insensitive accessible-name patterns for form submit buttons and headings so capitalization does not make the form locator empty; do not weaken actual required text assertions. Replace the unconditional fill method with a selected-field method whose field list comes from external JSON. TC01, TC02, and TC04 must still clearly assert the required confirmation control; TC03 must fill only the externally configured fields n

eeded to isolate email validity and must not fail first because confirmation is absent. Improve application-error discovery to include visible non-heading siblings immediately before the form as well as semantic alert/live/error candidates, without relying on exact undocumented error wording; placement above submit must still be asserted whenever such an application error exists. Keep data external, collision-safe email generation, no fixed sleeps, no inline test-case arrays/objects, and no test.skip/test.fail/conditional pass. Do not modify SUT source or weaken assertions to match the current implementation. Do not run tests, servers, package commands, Git commands, or commits. Summarize exact corrections.

Exact AI output

```text

Corrected exactly these four files:

- [fr01-registration.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr01-registration.json)
  - Added external `id`, `title`, and selected `fieldsToFill` data.
  - Configured TC03 to fill every required non-email-validity control, including confirmation.
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)
  - Typed case identity and field-selection properties.
- [registration.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/registration.page.ts)
  - Replaced registration/login `getByLabel` locators with anchored, escaped, case-insensitive label-to-container-to-input locators.
  - Added anchored case-insensitive accessible-name patterns for headings and submit buttons.
  - Replaced unconditional filling with externally driven selected-field filling.
- Expanded application-error discovery to semantic candidates and the visible non-heading sibling immediately preceding the form.
- [fr01-registration.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr01-registration.spec.ts)
  - Builds all four test titles from external JSON `id` and `title`.
  - Uses external field selections.
  - Explicitly asserts the required confirmation control in TC01, TC02, and TC04.
- Preserves all existing requirement assertions and error-placement checks.

No tests, servers, package commands, Git commands, or commits were run.

## Human review/action

Em đã đối chiếu correction với prompt và phát hiện email\_format\_partitions.field sToFill vẫn chứa confirmPassword, nên TC03 chưa cô lập đúng email validity. Em yêu cầu bỏ confirmPassword khỏi riêng field list của TC03, giữ nguyên confirmation assertions ở TC01, TC02 và TC04, không thay đổi requirement assertions hoặc source SUT. Chưa chạy test và chưa có defect claim.

# Interaction 0008

## AI tool

OpenAI Codex CLI 0.147.0 (model: gpt-5.6-sol)

## ISO date/time



2026-08-18T12:45:38Z

## Exact prompt

```text

Correct only tests/data/fr01-registration.json. In the email_format_partitions object, change fieldsToFill to exactly ["fullName", "email", "password"] so FR01-TC03 isolates email validity without depending on the missing confirmation field. Do not change any other JSON value, case ID, title, variant, assertion, test code, support code, page object, SUT source, or audit file. Do not run tests. Report the exact file and change made.

Exact AI output

```text

Updated [tests/data/fr01-registration.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr01-registration.json): changed `email\_format\_partitions.fieldsToFill` to exactly `["fullName", "email", "password"]`.

No other values changed. Tests were not run.

## Human review/action

Em đã kiểm tra toàn bộ execution output và final diff. Codex ban đầu sửa nhầm fieldsToFill của TC01 nhưng đã tự hoàn tác trước khi hoàn thành; final diff chỉ lỗi confirmationPassword khỏi email\_format\_partitions.fieldsToFill của TC03. Ba file TypeScript không thay đổi, TC01/TC02/TC04 vẫn giữ confirmation assertions, chưa chạy test, chưa có defect claim và source SUT không bị sửa.

# Interaction 0009

## AI tool

OpenAI Codex CLI 0.147.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-18T12:52:15Z

## Exact prompt

```text

Fix only tests/features/fr01-registration.spec.ts. Typecheck showed exactly four errors because Playwright 1.62.1 does not provide expect(locator).toBeRequired(). Replace exactly those four matcher calls with the supported assertion toHaveAttribute("required", ""), preserving each locator, custom assertion message, test ID, title, test flow, and the requirement that the control must be required. Do not change any other assertion or file. Do not modify test data, page objects, support code, SUT source, or audit files. Do not run tests. Report the exact replacements made.

Exact AI output

```text

Updated only [fr01-registration.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr01-registration.spec.ts).

Exact replacements:

- Lines 29, 67, and 189:

```
`expect(registration.confirmationPasswordInput).toBeRequired()`
→ `expect(registration.confirmationPasswordInput).toHaveAttribute("required",
```

""`

- Line 72:

```
`expect(input, ``Required ${field} control must be marked required``).toBeRequired()
→ `expect(input, ``Required ${field} control must be marked required``).toHaveAttribute("required", "")`
```

No tests were run. No other assertions or files were changed.

## Human review/action

Em đã review final diff và xác nhận Codex chỉ thay đúng bốn matcher toBeRequired không được Playwright 1.62.1 hỗ trợ bằng toHaveAttribute("required", ""), không làm yếu requirement assertion. FR01 JSON, page object và data loader không thay đổi. Typecheck đã thành công và Playwright liệt kê đúng bốn test FR01-TC01 đến FR01-TC04 trên Chromium. Chưa thực thi browser test, chưa có pass/fail hoặc defect claim và source SUT không bị sửa.

# Interaction 0010

## AI tool

OpenAI Codex CLI 0.147.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-20T05:36:32Z

## Exact prompt

````text

Extend the existing reviewed FR-01 Playwright automation with increment 2 covering exactly FR01-TC05 through FR01-TC08.

Read and preserve the existing design in:

- docs/hw04-traceability-test-matrix.md
- tests/data/fr01-registration.json
- tests/features/fr01-registration.spec.ts
- tests/pages/registration.page.ts
- tests/support/data-loader.ts
- README.md

Modify only these automation files when necessary:

- tests/data/fr01-registration.json
- tests/features/fr01-registration.spec.ts
- tests/pages/registration.page.ts
- tests/support/data-loader.ts

Do not modify frontend-web, frontend-admin, frontend-mobile, backend, documentation, evidence, configuration, package files, or ai-audit files. Do not run tests.

Requirements:

1. Add exactly four test declarations, with IDs and titles loaded from external JSON:

- FR01-TC05: password minimum 8 characters.
- FR01-TC06: password requires uppercase.
- FR01-TC07: password requires lowercase.
- FR01-TC08: password requires a digit.

2. Add external JSON objects with these exact keys:
 - password_length_boundary
 - password_missing_uppercase
 - password_missing_lowercase
 - password_missing_digit
3. Keep all test-case IDs, titles, inputs, variants, expected values, and fields ToFill in external JSON. Do not introduce inline test-case arrays or duplicate case data in the spec.
4. For every TC05-TC08 submission, fill exactly the existing controls:
["fullName", "email", "password"]
Do not depend on confirmPassword because these cases isolate password-rule behavior.
5. Use collision-safe unique email templates through the existing generateCollisionSafeEmail helper.
6. Use these password partitions so each case isolates its target rule:
 - TC05 seven-character invalid boundary: "Aa1!bbb"
 - TC05 eight-character valid boundary: "Aa1!bbbb"
 - TC06 missing uppercase: "abcdef1!"
 - TC07 missing lowercase: "ABCDEFG1!"
 - TC08 missing digit: "Abcdefg!"The exclamation mark is an allowed special character in FR-01.
7. TC05 must exercise both boundaries through the UI:
 - The 7-character value must remain on registration, show a visible non-empty application validation error above submit, and must not send POST /api/register
 - The 8-character value satisfies every specified password class and must reach the documented login destination. Capture registration requests without weakening this success oracle.
 - Do not assert an exact validation message because the matrix marks it EVIDENCE MISSING.
8. TC06-TC08 must each:
 - Submit through the UI.
 - Remain on the registration page.
 - Show a visible non-empty application validation error above submit without requiring exact wording.
 - Verify that POST /api/register was not sent, providing the state-delta evidence that the registration creation path was not invoked.
9. Add or reuse robust Page Object helpers for:
 - observing/counting POST /api/register requests;
 - locating a visible, non-empty application error within the registration view;
 - checking that the error is geometrically above the submit button.Keep locators form-scoped and compatible with the current label-parent-input DOM. Prefer web-first Playwright assertions. Do not use waitForTimeout, arbitrary sleeps, brittle CSS chains, getByLabel for the current unlabeled DOM, or exact browser validation wording.
10. Preserve FR01-TC01 through FR01-TC04 unchanged in intent and strength. Do not remove, skip, conditionally bypass, soften, or reorder their requirement assertions merely because the current SUT fails them.
11. Keep TypeScript strict-compatible and Playwright 1.62.1-compatible. Do not use unsupported matchers such as toBeRequired.
12. Do not change the SUT to make tests pass. Do not claim any pass/fail result

or defect because no execution is authorized in this interaction.

After editing, report:

- exact files changed;
- the four JSON keys and four generated test IDs;
- the assertion/oracle used by each case;
- confirmation that no tests were run and no SUT or audit file was modified.

Exact AI output

````text

Implemented FR-01 increment 2 with exactly four new test declarations.

Changed files:

- [fr01-registration.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr01-registration.json:211)
- [fr01-registration.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr01-registration.spec.ts:287)
- [registration.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/registration.page.ts:103)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts:22)

JSON keys and generated IDs:

- `password\_length\_boundary` → `FR01-TC05`
- `password\_missing\_uppercase` → `FR01-TC06`
- `password\_missing\_lowercase` → `FR01-TC07`
- `password\_missing\_digit` → `FR01-TC08`

Oracles:

- TC05: seven-character value remains on registration, shows a visible non-empty application error above submit, and records zero registration POSTs. Eight-character value records one POST and must expose the documented login heading, form, email, and password controls.
- TC06: remains on registration, visible non-empty error above submit, zero registration POSTs.
- TC07: same rejection and zero-POST oracle.
- TC08: same rejection and zero-POST oracle.

All submissions fill only `fullName`, `email`, and `password`, using collision-safe email templates. TC01-TC04 assertions remain intact.

No Playwright tests or SUT were run. Static TypeScript, JSON syntax, and diff checks completed successfully. No SUT, documentation, configuration, package, evidence, or audit file was modified; the pre-existing untracked audit files were left untouched.

````

Human review/action

Em đã review toàn bộ bốn file automation và xác nhận increment 2 bổ sung đúng FR01-TC05 đến FR01-TC08. ID, title, input, fieldsToFill và expected values được lấy từ external JSON; mỗi case chỉ điền fullName, email và password. Các password partition cô lập đúng boundary 7/8 ký tự, thiếu chữ hoa, thiếu chữ thường và thiếu chữ số. Invalid cases giữ oracle lỗi ứng dụng không rỗng nằm trên submit cùng zero POST /api/register; boundary 8 giữ success oracle tới trang đăng nhập và một registration POST. TC01-TC04 không bị làm yếu. Typecheck thành công và Playwright liệt kê đúng 8 test FR01 trên Chromium. Chưa thực thi browser test, chưa có pass/fail hoặc defect claim và source SUT không bị sửa.

Interaction 0011

AI tool

OpenAI Codex CLI 0.147.0 (model: gpt-5.6-sol)

ISO date/time
2026-08-20T06:35:45Z

Exact prompt

```text

Extend the existing reviewed FR-01 Playwright automation with increment 3 covering exactly FR01-TC09 through FR01-TC12.

Read and preserve:

- docs/hw04-traceability-test-matrix.md
- README.md
- api\_specification.md
- playwright.config.ts
- .env.example
- tests/data/fr01-registration.json
- tests/features/fr01-registration.spec.ts
- tests/pages/registration.page.ts
- tests/support/data-loader.ts

Modify only these automation files when necessary:

- tests/data/fr01-registration.json
- tests/features/fr01-registration.spec.ts
- tests/pages/registration.page.ts
- tests/support/data-loader.ts

Do not modify SUT source, documentation, configuration, package files, evidence, or ai-audit files. Do not run tests.

Add exactly four test declarations whose IDs and titles come from external JSON:

- FR01-TC09: password requires an allowed special character.
- FR01-TC10: each of @, \$, !, %, \*, ?, & satisfies the special-character rule.
- FR01-TC11: confirmation equal-versus-unequal decision table and error placement.
- FR01-TC12: unique-email registration followed by duplicate registration through the API.

Add these exact external JSON keys:

- password\_missing\_allowed\_special
- password\_each\_allowed\_special
- confirmation\_match\_matrix
- api\_unique\_then\_duplicate

General rules:

1. Keep IDs, titles, row names, inputs, fieldsToFill, endpoints, expected statuses/messages/counts, symbols, and decision-table values in external JSON. Do not add inline case arrays.
2. Continue using collision-safe run-unique email templates.
3. Preserve FR01-TC01 through FR01-TC08 unchanged in intent and assertion strength.
4. Keep current form-scoped label-parent-input locators. Do not use getByLabel for the current DOM.
5. Do not use waitForTimeout, arbitrary sleeps, test.skip, test.fixme, conditionally bypassed requirement assertions, exact browser-native wording, or unsupported

ed Playwright matchers.

6. Exact application validation wording remains EVIDENCE MISSING and must not be asserted.

7. Do not change the SUT to make tests pass.

TC09:

- Use password "Abcdefg1": eight characters with uppercase, lowercase, and digit, but none of the seven permitted special characters.
- Fill exactly ["fullName", "email", "password"].
- Submit through the UI.
- Assert it remains on registration, displays a visible non-empty application error above submit, and sends zero POST /api/register requests.
- Reuse the reviewed rejection helper where appropriate.

TC10:

- Store exactly seven JSON rows, one for each symbol: @, \$, !, %, \*, ?, &.
- Vary only the permitted symbol in these passwords:  
"Abcdef1@", "Abcdef1\$", "Abcdef1!", "Abcdef1%",  
"Abcdef1\*", "Abcdef1?", "Abcdef1&".
- Fill exactly ["fullName", "email", "password"].
- Every row must use a distinct collision-safe email.
- Every row must submit through the UI, reach the documented login destination, and send exactly one POST /api/register.
- Use test.step for readable row evidence.
- Use expect.soft only within this seven-row test so all listed symbols are exercised in one test declaration even when one or more rows fail. Soft assertions must retain the full navigation and request-count requirements.
- Use test.slow() rather than reducing assertion timeouts, so all seven rows can complete.
- Do not test unlisted special characters.
- Runtime users created by successful rows will be removed by restoring the controlled database backup after execution; do not invent a cleanup endpoint.

TC11:

- Store exactly two JSON rows: unequal and equal.
- Use a specification-valid password such as "Abcdef1!".
- Unequal row confirmation must differ while remaining a valid password value.
- Equal row confirmation must equal the password.
- Fill exactly ["fullName", "email", "password", "confirmationPassword"].
- First assert the confirmation-password control is visible, required, and type=password as a hard prerequisite.
- Unequal row must remain on registration, show a visible non-empty application error above submit, and send zero POST /api/register.
- Equal row must not be rejected for mismatch: it must reach the documented login destination and send exactly one POST /api/register.
- Any visible application error asserted for the unequal row must be geometrically above submit.
- Do not weaken or bypass the prerequisite if the current SUT lacks the control.
- Any user created by the equal row will be removed by restoring the controlled database backup after execution.

TC12:

- Use Playwright's request fixture and an absolute API URL composed from:  
process.env.PW\_API\_URL ?? "http://localhost:3000"  
plus the endpoint stored in JSON.
- Do not accidentally send TC12 to the frontend baseUrl.
- JSON must contain a first valid registration record and a second record that reuses the same generated email.
- The API body must follow the documented contract and omit confirmationPassword.
- First POST must return the documented 200 status, documented success message, and a numeric id.

- Capture the first id.
- Second POST with the same email must be rejected. Because duplicate failure status/body are EVIDENCE MISSING, assert that the second response is not successful without inventing an exact status or wording.
- Preserve response bodies sufficiently for clear assertion diagnostics.
- This test must demonstrate the unique-then-duplicate state transition; do not mock or intercept the API.
- Runtime records will be removed by restoring the controlled database backup after execution; do not invent a cleanup endpoint.

Keep TypeScript strict-compatible and Playwright 1.62.1-compatible. The final file must contain exactly twelve FR01 test declarations total.

After editing, report:

- exact files changed;
- the four added JSON keys and IDs;
- the seven TC10 symbols;
- the TC11 decision-table rows;
- the TC12 API transition oracle;
- confirmation that no tests were run and no SUT or audit file was modified.

## Exact AI output

```text

Implemented FR-01 increment 3 with exactly 12 total test declarations.

Changed files:

- [fr01-registration.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr01-registration.json)
- [fr01-registration.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr01-registration.spec.ts)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)

Added JSON keys and IDs:

- `password\_missing\_allowed\_special` - `FR01-TC09`
- `password\_each\_allowed\_special` - `FR01-TC10`
- `confirmation\_match\_matrix` - `FR01-TC11`
- `api\_unique\_then\_duplicate` - `FR01-TC12`

TC10 symbols: `@`, `\$`, `!`, `%`, `\*`, `?`, `&`.

TC11 decision table:

- `unequal`: `Abcdef1!` / `Abcdef1?` → remain on registration, application error above submit, zero registration POSTs.
- `equal`: `Abcdef1!` / `Abcdef1!` → reach login, exactly one registration POST.

TC12 oracle:

- First unique-email POST → status `200`, documented success message, numeric `id`.
- Capture that `id`.
- Second POST reusing the generated email → `response.ok()` is false, without asserting undocumented status or wording.

Static JSON, diff, and strict TypeScript checks passed. No tests were run. No SUT, documentation, configuration, package, evidence, page-object, or audit file was modified; existing untracked audit artifacts were left untouched.

````

## ## Human review/action

Em đã review toàn bộ increment 3 và xác nhận automation bổ sung đúng FR01-TC09 đến FR01-TC12, nâng tổng FR-01 lên đúng 12 test. TC09 cô lập trường hợp thiếu allowed special; TC10 dùng bày external JSON rows cho @, \$, !, %, \*, ?, & với test.step, test.slow và soft assertions giữ nguyên navigation cùng one-POST oracle cho từng row; TC11 giữ confirmation control visible, required và type=password làm hard prerequisite trước decision table unequal/equal; TC12 dùng absolute PW\_API\_URL, body đúng API contract, kiểm tra first 200/success message/numeric id rồi yêu cầu duplicate response không successful mà không suy diễn status hoặc wording. TC01-TC08 không bị làm yếu, Page Object không cần sửa. Typecheck thành công và Playwright liệt kê đúng FR01-TC09 đến FR01-TC12 trên Chromium. Chưa thực thi browser/API test, chưa có pass/fail hoặc defect claim và source SUT không bị sửa.

## # Interaction 0012

## ## AI tool

OpenAI Codex CLI 0.147.0 (model: gpt-5.6-sol)

## ## ISO date/time

2026-08-20T08:53:09.072Z

## ## Exact prompt

````text

Extend the reviewed Playwright automation with FR-07 increment 1 covering exactly FR07-TC01 through FR07-TC04.

Read and preserve:

- README.md
- api_specification.md
- docs/hw04-traceability-test-matrix.md
- playwright.config.ts
- tests/support/data-loader.ts
- the existing FR-01 automation
- frontend-web/src/App.jsx
- frontend-web/src/context/CartContext.jsx
- frontend-web/src/pages/Home.jsx
- frontend-web/src/pages/ProductDetail.jsx
- frontend-web/src/pages/Cart.jsx
- backend/server.js
- backend/database.js

Create or modify only these automation files when necessary:

- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/catalog.page.ts
- tests/pages/cart.page.ts
- tests/support/data-loader.ts

Do not modify FR-01 data, specs, or page objects. Do not modify SUT source, documentation, configuration, packages, evidence, database files, or ai-audit files. Do not run Playwright tests and do not start the SUT. Static JSON parsing and TypeScript typechecking are allowed.

Add exactly four test declarations. IDs and titles must come from external JSON:

- FR07-TC01 populated cart structure, currency, and one descriptive h1.
- FR07-TC02 invalid cart API authentication partitions.
- FR07-TC03 adding the same product twice produces one row with quantity two.

- FR07-TC04 the plus control increments only the selected row and recalculates totals.

Add these exact external JSON keys:

- populated_cart_two_products
- cart_invalid_authentication
- same_product_ui_addition
- quantity_increment

Keep IDs, titles, paths, endpoints, product identities, expected headings, headers, quantities, prices, currency rules, token partitions, request bodies, login credentials, and expected counts in external JSON. Do not add inline case arrays or duplicate case data in the spec.

UI fixture rules:

- The web cart is React in-memory state and is not connected to the backend cart API.
- Build UI cart state only through visible product-card actions on the home page within the current browser context.
- Do not inject React state, use local storage, call private component functions, or use the backend cart API to seed UI cases.
- Scope each product card by its visible product heading, then locate its add-to-cart button inside that card.
- Navigate to the cart through the visible Giỏ hàng navigation link.
- Keep locators semantic and page-object based. Do not use brittle CSS chains, nth-based product selection, arbitrary sleeps, or waitForTimeout.
- Each Playwright test receives a fresh browser context, so UI cart state is isolated without database mutation.

TC01 requirements:

- Seed exactly two distinct products by activating each selected home product card add action once.
- Assert exactly two product rows.
- Assert exactly one descriptive h1 for the cart.
- Assert the required column labels: Sản phẩm, Đơn giá, Số lượng, Thành tiền, and Thao tác.
- Assert each row exposes plus and minus quantity controls.
- Assert displayed unit prices and line totals contain đ, contain a thousands separator, and numerically match the external product prices and quantities.
- Use expect.soft where useful so independent structural requirements are all evaluated without weakening any assertion.

TC02 requirements:

- Use the Playwright request fixture and an absolute API URL composed from process.env.PW_API_URL ?? "http://localhost:3000" plus endpoints stored in JSON.
- Store exactly three invalid-authentication rows in JSON: missing token, malformed token, and expired token.
- The expired token must be a real JWT fixture signed for the current SUT with a n exp value in the past, stored as a token literal in external JSON. Do not expose the signing secret in the spec.
- Obtain a valid baseline token by logging in with the seeded dedicated user credentials from external JSON.
- Snapshot that users cart with authenticated GET /api/cart before the invalid attempts.
- For every invalid-authentication row, exercise both GET /api/cart and POST /api/cart using test.step.
- Assert each invalid request is not successful without inventing an exact status or response wording. Preserve response status and body in assertion diagnostics.
- Use soft assertions for the six rejection responses so all authentication partitions execute.
- Fetch the cart again with the valid token and assert deep equality with the baseline.

seline snapshot, proving rejected POST attempts caused no mutation.
- Do not mock or intercept the API.

TC03 requirements:

- Start with a fresh empty UI cart.
- Activate the same home product cards add action exactly twice.
- Open the cart and assert exactly one row for that product, quantity 2, line total equal to twice the external unit price, and cart total equal to the sum of all displayed line totals.
- Do not compensate for or retry failed add actions. The two activations are the tested state transition.
- After asserting row count, use a deterministic row-scoped locator for the remaining assertions without conditionally bypassing them.

TC04 requirements:

- Seed one target product through one visible home product-card add action, then open the cart.
- Capture row count, selected-row quantity, selected-row line total, and cart total before interaction.
- Require a row-scoped button whose accessible name is exactly + as a hard prerequisite.
- Click it once.
- Assert only the selected row quantity changes from n to n+1, its line total changes consistently, total row count is unchanged, and the cart total is recalculated consistently.
- Do not skip, soften, synthesize, or bypass the plus-control requirement if the current SUT does not provide it.

Page-object and calculation rules:

- Use role-based table, row, columnheader, cell, button, link, and heading locators where available.
- Scope cart row locators by external product name.
- Provide helpers that parse displayed currency by removing presentation separators while retaining a numeric comparison.
- Verify a displayed thousands separator without requiring one locale-specific separator character.
- Provide helpers for unit price, quantity, line amount, cart total, and summing displayed line amounts.
- Keep TypeScript strict-compatible and Playwright 1.62.1-compatible.
- Do not use unsupported matchers such as toBeRequired.

Preserve every existing FR01-TC01 through FR01-TC12 declaration and assertion unchanged in intent and strength.

Do not claim pass, failure, or defect because no execution is authorized.

After editing, report:

- exact files changed;
- the four JSON keys and generated test IDs;
- how each UI fixture is built;
- the assertion and state-delta oracle for each case;
- confirmation that no Playwright tests ran and no SUT, database, evidence, configuration, documentation, package, or audit file was modified.

Exact AI output

```text

Implemented FR-07 increment 1 with exactly four test declarations.

#### Changed files:

- [fr07-cart.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr07-cart.json)

- [fr07-cart.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr07-cart.spec.ts)
- [catalog.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/catalog.page.ts)
- [cart.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/cart.page.ts)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)

External JSON keys and generated IDs:

- `populated\_cart\_two\_products` → `FR07-TC01`
- `cart\_invalid\_authentication` → `FR07-TC02`
- `same\_product\_ui\_addition` → `FR07-TC03`
- `quantity\_increment` → `FR07-TC04`

Fixture construction and oracles:

- TC01: clicks two distinct heading-scoped home product cards once each, then uses the visible cart link. Verifies two rows, one descriptive `h1`, required headers, row-scoped `+/-` controls, quantities, unit prices, line totals, `₫`, and locale-independent thousands separators.
- TC02: logs in as the external seeded user, snapshots authenticated `/api/cart`, exercises GET and POST for missing, malformed, and genuinely expired JWT partitions, records status/body diagnostics, and deep-compares the final authenticated cart with the baseline.
- TC03: clicks the same heading-scoped product card exactly twice from a fresh context. Verifies one product row, quantity two, doubled line amount, and cart total equal to all displayed line amounts.
- TC04: adds one target product once, requires an exact row-scoped `+` button, captures pre-interaction row/quantity/line/total state, clicks once, and verifies the `n → n+1` quantity and corresponding line/sum/cart-total deltas without changing row count.

Static validation completed successfully:

- JSON parsing and structural checks
- Expired JWT signature/expiry verification
- Strict TypeScript typechecking
- Exactly four FR-07 test declarations
- No FR-01 diff

No Playwright tests ran and no SUT was started. No SUT, database, evidence, configuration, documentation, package, or audit file was modified. The two pre-existing untracked `ai-audit` files remained untouched. No pass, failure, or defect result is claimed.

## Human review/action

Em đã review toàn bộ năm file automation và xác nhận FR-07 increment 1 bổ sung đúng FR07-TC01 đến FR07-TC04. ID, title, product fixture, authentication partitions, paths, labels, prices, quantities và expected counts đều lấy từ external JSON. TC01 tạo cart qua hai product card riêng biệt và giữ đầy đủ structure, h1, column, quantity-control cùng currency assertions. TC02 dùng login token hợp lệ để snapshot cart, thực hiện đủ missing, malformed và JWT đã ký nhưng hết hạn qua cả GET và POST, sau đó so sánh cart trước/sau mà không suy diễn status hoặc wording. TC03 kích hoạt cùng product card đúng hai lần, không retry hoặc bù hành vi. TC04 giữ nút + row-scoped làm hard prerequisite trước state-delta assertions. FR01-TC01 đến FR01-TC12 không thay đổi. Typecheck thành công và Playwright liệt kê đúng bốn test FR07-TC01 đến FR07-TC04 trên Chromium. Chưa thực thi browser/API test, chưa có pass/fail hoặc defect claim và source SUT không bị sửa.

# Interaction 0013

## AI tool

OpenAI Codex CLI 0.147.0 (model: gpt-5.6-sol)

## ISO date/time  
2026-08-20T09:49:04.512Z

## Exact prompt

```text

Correct only tests/features/fr07-cart.spec.ts after the first FR-07 increment 1 Chromium development run.

Do not modify JSON data, page objects, support code, FR-01 automation, SUT source, database, evidence, configuration, documentation, packages, or ai-audit files. Do not run Playwright tests and do not start or stop the SUT. TypeScript typechecking is allowed.

Make exactly these two corrections:

1. FR07-TC01 accumulated multiple independent soft missing-locator assertions and reached the default 30-second test timeout before every soft assertion finished. Add test.slow() inside FR07-TC01 so Playwright provides the standard extended test budget. Preserve every existing soft assertion, locator, expected count, ID, title, fixture action, and requirement oracle. Do not reduce assertion timeouts, remove assertions, convert failures to warnings, or otherwise weaken the test.
2. FR07-TC04 used an immediate locator.count() followed by a synchronous expect and observed zero rows even though the failure DOM snapshot already contained the seeded AirPods row. Replace the initial row-count check with a web-first await expect(cart.dataRows()).toHaveLength(testCase.expected.counts.rows), then capture beforeRowCount from the same locator for the later unchanged-row-count comparison. Preserve the product-row assertion, exact row-scoped plus-button hard prerequisite, quantities, prices, line totals, cart-total assertions, and every post-click state-delta oracle.

Do not change any other line or file. Do not claim a pass, failure, or defect because no corrected execution is authorized in this interaction.

After editing, report the exact file changed, the exact two corrections, confirm that all four FR07 test declarations and all existing assertions remain, and confirmation that no Playwright test or SUT was run.

Exact AI output

```text

Changed only [tests/features/fr07-cart.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr07-cart.spec.ts).

- FR07-TC01: added `test.slow()` inside the test.
- FR07-TC04: added a shared `dataRows` locator, awaited `toHaveLength(...)` , then captured `beforeRowCount` from that locator.

All four FR07 test declarations and all existing assertions remain. TypeScript typechecking completed without diagnostics. No Playwright test was run, and the SUT was neither started nor stopped.

## ## Human review/action

Em đã review final diff và xác nhận Codex chỉ sửa tests/features/fr07-cart.spec.ts tại đúng hai điểm. FR07-TC01 thêm test.slow để toàn bộ independent soft assertions có đủ test budget mà không giảm timeout hoặc làm yếu oracle. FR07-TC04 thay immediate count bằng web-first toHaveCount trước khi ghi nhận beforeRowCount, khắc phục timing issue đã được failure DOM chứng minh trong khi giữ nguyên hard prerequisite nút + và toàn bộ state-delta assertions. Cả bốn FR07 test declarations và các assertion hiện có vẫn được giữ nguyên. Typecheck thành công, chưa rerun Playwright, chưa có kết quả corrected run hoặc defect claim mới và source SUT không bị sửa.

## # Interaction 0014

## ## AI tool

OpenAI Codex CLI 0.148.0 (model: gpt-5.6-sol). The initial Codex CLI 0.147.0 attempt was aborted by a model-cache incompatibility; the user authorized updating the CLI before the successful final execution.

## ## ISO date/time

2026-08-20T10:49:08Z

## ## Exact prompt

```text

Extend the existing reviewed FR-07 Playwright automation with increment 2 covering exactly FR07-TC05 through FR07-TC08.

Read and preserve:

- docs/hw04-traceability-test-matrix.md
- README.md
- playwright.config.ts
- frontend-web/src/App.jsx
- frontend-web/src/context/CartContext.jsx
- frontend-web/src/pages/Home.jsx
- frontend-web/src/pages/ProductDetail.jsx
- frontend-web/src/pages/Cart.jsx
- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/support/data-loader.ts

Modify only these automation files when necessary:

- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/pages/product-detail.page.ts
- tests/support/data-loader.ts

Do not modify SUT source, FR-01 automation, documentation, configuration, package files, evidence, database files, or ai-audit files. Do not run Playwright test S.

Add exactly four test declarations whose IDs and titles come from external JSON:

- FR07-TC05: decrement quantity from 2 to 1 and recalculate totals.
- FR07-TC06: dangerous delete action is red and opens confirmation before removal.
- FR07-TC07: cancelling deletion preserves the complete cart state.

- FR07-TC08: confirming deletion removes only the selected row and updates total and cart badge.

Add these exact external JSON keys:

- quantity_decrement_above_minimum
- deletable_item
- deletion_cancel
- deletion_confirm

General rules:

1. Keep IDs, titles, paths, product identities, prices, quantities, labels, action semantics, expected counts, expected totals, and fixture values in external JSON. Do not add inline test-case arrays or duplicate case data in the spec.
2. Build all UI cart fixtures through visible user interactions in a fresh Playwright context. Do not mutate React internals, inject cart state, call implementation functions, mock cart behavior, or modify the SUT.
3. Preserve FR07-TC01 through FR07-TC04 unchanged in intent and assertion strength, including the reviewed interaction-0013 stabilization.
4. Preserve FR01-TC01 through FR01-TC12 unchanged.
5. Keep locators page- and row-scoped. Prefer accessible roles and names. Avoid brittle CSS chains, positional selectors when semantic scoping is available, and implementation-only selectors.
6. Do not use `waitForTimeout`, arbitrary sleeps, `test.skip`, `test.fixme`, retries, compensating duplicate clicks, conditional requirement bypasses, or unsupported Playwright matchers.
7. Exact dialog, cancel, confirmation, toast, and success wording is EVIDENCE MISSING. Do not assert invented exact wording.
8. Use web-first assertions before reading numeric or geometric state.
9. Do not change the SUT to make tests pass and do not claim pass/fail or a defect because no execution is authorized in this interaction.

TC05 – quantity_decrement_above_minimum:

- Construct a two-product cart so “only the selected row changes” is observable.
- For the target product, use its visible Product Detail quantity control to enter 2 and activate Add to cart exactly once.
- Do not double-click, retry, or compensate if the current SUT ignores the activation.
- Add one distinct comparison product once through the visible UI.
- Navigate through the visible cart link.
- Hard prerequisites must establish exactly two rows, exactly one target row at quantity 2, exactly one comparison row at quantity 1, and exactly one row-scoped control for the target.
- Capture row count, both quantities, both line amounts, and displayed cart total only after web-first prerequisites.
- Click the target row’s control exactly once.
- Assert row count is unchanged; target quantity changes 2 to 1; target line amount decreases by one target unit price; comparison quantity and line amount are unchanged; cart total decreases by one target unit price and equals the sum of displayed line amounts.
- Do not infer behavior at quantity 1.

TC06 – deletable_item:

- Add exactly one externally defined product once through the visible Home UI and open the cart.
- Hard prerequisites must establish the target row and its row-scoped delete action.
- Assess the computed dangerous-action color semantically as red without asserting an exact shade.
- Capture row count, target quantity, target line amount, and cart total.
- Activate Delete exactly once without confirming.
- Require a visible accessible confirmation dialog before deletion.
- While the dialog awaits a decision, assert the target row and captured cart state.

ate remain unchanged.

- Close or cancel the dialog after assertions so this case intentionally performs no deletion.

TC07 – deletion_cancel:

- Add exactly one externally defined product once through the visible Home UI and open the cart.
- Capture the complete baseline: row count, target row, quantity, line amount, displayed cart total, and sum of displayed line amounts.
- Open the accessible confirmation dialog through the target row's delete action.
- Locate the semantic cancel/dismiss action supplied by the dialog without asserting one invented exact label. Keep any locator vocabulary or action semantics in external JSON.
- Activate cancel/dismiss once.
- Assert the dialog closes and the complete captured cart state is unchanged.

TC08 – deletion_confirm:

- Add two distinct externally defined products once each through the visible Home UI and open the cart.
- Capture both row states, deleted-row line amount, remaining-row state, row count, and displayed cart total.
- Open the accessible confirmation dialog from the selected row.
- Locate the semantic confirm/destructive action without asserting invented exact dialog wording. Keep locator vocabulary or action semantics in external JSON.
- Confirm exactly once.
- Assert the dialog closes, only the selected row disappears, the other row quantity and line amount remain unchanged, row count decreases by one, displayed cart total decreases by the deleted line amount, and the resulting total equals the sum of remaining displayed line amounts.
- Assert the cart navbar badge is visible and reflects the externally defined remaining cart count. Do not invent a badge or silently skip this requirement if the current SUT lacks one.

Page Object expectations:

- Reuse current CartPage and CatalogPage behavior where appropriate.
- Add a ProductDetailPage only if needed for TC05's visible quantity fixture.
- Add robust helpers for a row-scoped delete action, accessible confirmation dialog, semantic cancel/confirm actions, dangerous red-color assessment, cart badge lookup/parsing, and stable cart-state reads.
- Keep product-detail quantity and Add controls scoped to the displayed product view.
- Prefer Playwright web-first assertions in the spec; Page Object methods may read values only after those prerequisites.

Keep TypeScript strict-compatible and Playwright 1.62.1-compatible. The final FR07 spec must contain exactly eight FR07 test declarations total.

After editing, report:

- exact files changed;
- the four added JSON keys and generated IDs;
- the TC05 two-product fixture and state-delta oracle;
- the TC06 dialog-before-delete oracle;
- the TC07 cancel/no-mutation oracle;
- the TC08 confirmed-delete, total, and badge oracle;
- static validation performed;
- confirmation that no tests were run and no SUT, FR-01, audit, evidence, database, configuration, documentation, or package file was modified.

...

Exact AI output

```text

Implemented FR-07 increment 2 with exactly eight total FR-07 test declarations.

Changed automation files:

- [fr07-cart.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr07-cart.json)
- [fr07-cart.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr07-cart.spec.ts)
- [cart.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/cart.page.ts)
- [catalog.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/catalog.page.ts)
- [product-detail.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/product-detail.page.ts)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)

Added JSON records:

- `quantity\_decrement\_above\_minimum` → `FR07-TC05`
- `deletable\_item` → `FR07-TC06`
- `deletion\_cancel` → `FR07-TC07`
- `deletion\_confirm` → `FR07-TC08`

Oracles implemented:

- TC05: AirPods quantity 2 plus one Keychron comparison row; verifies rows remain 2, AirPods quantity `2 → 1`, line amount `12,000,000 → 6,000,000`, Keychron remains quantity 1/`4,000,000`, and total `16,000,000 → 10,000,000`.
- TC06: verifies the iPhone delete action is semantically red, deletion does not happen before an accessible confirmation dialog appears, and the full `30,000,000` cart state remains unchanged while awaiting—and after dismissing—the dialog.
- TC07: captures the complete one-row Samsung baseline and verifies dialog cancellation preserves row count, identity, quantity, line amount, line sum, and `28,000,000` total.
- TC08: deletes only AirPods from an AirPods/Keychron cart, verifies rows `2 → 1`, total `10,000,000 → 4,000,000`, unchanged Keychron state, remaining-line sum consistency, and a visible cart badge count of `1`.

Static validation completed:

- `npm run typecheck` passed.
- JSON parsing and fixture arithmetic invariants passed.
- Confirmed exactly eight FR-07 declarations and one declaration for each new JS ON-driven case.
- Confirmed TC01-TC04 implementation body remains byte-for-byte unchanged.
- `git diff --check` passed.
- Forbidden-pattern scan found no sleeps, skips, fixmes, retries, duplicate-click helpers, storage injection, or React cart-internal access.

No Playwright tests were run, and no execution result or SUT defect is claimed. No SUT, FR-01, audit, evidence, database, configuration, documentation, or package file was modified. The two already-untracked `ai-audit` files present at the start were left untouched.

## Human review/action

Từ chối output 0014. Human action: Em đã review increment 2 và phát hiện TC05-TC08 dùng exact label “Tổng cộng” để định vị cart total trước target action. Điều



này khiến TC06-TC08 có thể bị chặn bởi requirement của FR07-TC10 trước khi dialog/delete oracles được thực thi, nên các case chưa được cô lập đúng. TC05 vẫn phải giữ visible quantity fixture và đúng một Add click, không được bù hành vi nếu fixture thất bại. Cần lưu output hiện tại làm rejected artifact và sửa locator cart summary để không khẳng định exact label; FR07-TC10 sẽ giữ exact-label oracle. Chưa chạy Playwright và SUT không bị sửa.

# Interaction 0015

## AI tool

OpenAI Codex CLI 0.148.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-20T11:20:40Z

## Exact prompt

```text

Correct the rejected FR-07 increment 2 proposal for exactly FR07-TC05 through FR07-TC08.

Read and preserve:

- docs/hw04-traceability-test-matrix.md
- README.md
- playwright.config.ts
- frontend-web/src/App.jsx
- frontend-web/src/context/CartContext.jsx
- frontend-web/src/pages/Home.jsx
- frontend-web/src/pages/ProductDetail.jsx
- frontend-web/src/pages/Cart.jsx
- ai-audit/prompts/0014-fr07-cart-controls-delete.txt
- ai-audit/artifacts/0014-rejected-fr07-increment2/fr07-cart.json
- ai-audit/artifacts/0014-rejected-fr07-increment2/fr07-cart.spec.ts
- ai-audit/artifacts/0014-rejected-fr07-increment2/cart.page.ts
- ai-audit/artifacts/0014-rejected-fr07-increment2/catalog.page.ts
- ai-audit/artifacts/0014-rejected-fr07-increment2/product-detail.page.ts
- ai-audit/artifacts/0014-rejected-fr07-increment2/data-loader.ts
- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/pages/product-detail.page.ts
- tests/support/data-loader.ts

The 0014 proposal was rejected because TC05-TC08 used the exact required label "Tổng cộng" to locate the displayed cart total before their target actions. That exact-label requirement belongs to FR07-TC10. On a SUT using another label, TC06-TC08 would stop before exercising their dialog, cancel, confirm, and badge oracles.

Modify only these automation files when necessary:

- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/pages/product-detail.page.ts
- tests/support/data-loader.ts

Do not modify SUT source, FR-01 automation, documentation, configuration, packaging,

e files, database, evidence, or ai-audit files. Do not run browser or API tests.

Correction requirements:

1. Preserve exactly eight FR07 test declarations total and exactly one declaration for each FR07-TC05 through FR07-TC08. Keep IDs, titles, fixtures, prices, quantities, expected numeric totals, expected row counts, dialog action vocabulary, and badge counts in external JSON.
2. Preserve FR07-TC01 through FR07-TC04 unchanged in intent and assertion strength.
3. TC05-TC08 must read and compare the displayed cart summary amount without asserting or requiring an exact summary label. Do not use `expected.totalLabel`, "Tổng cộng", or "Tổng tạm tính" as the prerequisite locator for these four tests.
4. Keep the summary locator scoped to the cart view and distinguish the summary amount from table line amounts. Avoid brittle CSS chains and arbitrary positional selectors. Prefer web-first assertions before Page Object value reads.
5. Do not encode "Tổng tạm tính" as accepted expected behavior. FR07-TC10 will independently assert that the exact required label is "Tổng cộng".
6. TC05 must keep its visible Product Detail quantity fixture, quantity 2, and exactly one Add-to-cart activation. Do not retry, double-click, compensate, inject storage, or access React internals. If that visible fixture does not create the required cart state, its decrement action and later assertions must remain EVI DENCE NOT REACHED rather than being bypassed.
7. TC05 must retain the two-product fixture and hard prerequisites, then require the row-scoped minus control and verify only the target changes from quantity 2 to 1, its line amount halves, the comparison row remains unchanged, row count remains unchanged, and displayed cart total equals the remaining line sum and expected numeric delta.
8. TC06 must retain the red dangerous-action assessment and require an accessible dialog before deletion. The selected row and complete numeric cart state must remain unchanged while the dialog awaits a decision and after dismissing it.
9. TC07 must retain semantic cancel/dismiss discovery and verify the dialog closes while row identity, quantity, line amount, line sum, and displayed cart total remain unchanged.
10. TC08 must retain semantic confirm discovery and verify only the selected row disappears after confirmation, the comparison row remains unchanged, displayed cart total equals the remaining line sum and expected numeric total, and the visible cart badge has the expected count.
11. Keep row-scoped controls, accessible dialog roles, semantic dialog actions, dangerous red-color assessment, cart badge behavior, and stable state reads in Page Objects. Keep TypeScript strict-compatible and Playwright 1.62.1-compatible.
12. Do not use `waitForTimeout`, arbitrary sleeps, `test.skip`, `test.fixme`, conditional requirement bypasses, retries that compensate for SUT behavior, `getByLabel` for unlabeled DOM, exact dialog wording, unsupported matchers, or direct SUT-state injection.

Perform static validation only:

- parse the JSON and verify fixture arithmetic;
- run `npm run typecheck`;
- list the four increment-2 tests without executing them;
- confirm exactly eight FR07 declarations;

- confirm forbidden patterns are absent;
- confirm no SUT, FR-01, audit, evidence, database, configuration, documentation, or package file changed.

After editing, report:

- exact automation files changed relative to the current rejected proposal;
- how the cart summary amount is located without asserting its label;
- confirmation that TC05 still performs exactly one visible Add activation;
- the retained TC05-TC08 oracles;
- static validation performed;
- confirmation that no tests were run and no prohibited file was modified.

...

Exact AI output

```text

Corrected the rejected FR-07 increment 2 proposal.

Changed relative to the 0014 artifact:

- [tests/data/fr07-cart.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr07-cart.json:182)
- [tests/features/fr07-cart.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr07-cart.spec.ts:452)
- [tests/pages/cart.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/cart.page.ts:334)
- [tests/support/data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts:359)

`catalog.page.ts` and `product-detail.page.ts` remain byte-identical to the rejected proposal.

The summary amount is now located inside the cart's semantic `main` view, using a currency-only displayed value explicitly excluded from table descendants. Thus line amounts cannot satisfy the locator, and TC05-TC08 do not reference `expected.totalLabel`, `Tổng cộng`, or `Tổng tạm tính`. Web-first visibility and amount assertions precede numeric Page Object reads.

Retained behavior:

- TC05 performs exactly one visible Product Detail Add activation. Its two-row, target-quantity-2, comparison-quantity-1, and row-scoped-minus prerequisites remain hard failures. It verifies target quantity/line halving, unchanged comparison row, unchanged row count, line-sum consistency, expected total, and numeric delta.
- TC06 assesses the delete action as semantically red, requires an accessible dialog, and preserves the complete numeric state while awaiting a decision and after dismissal.
- TC07 discovers cancel/dismiss semantically and verifies dialog closure plus unchanged row identity, quantity, line amount, line sum, and total.
- TC08 discovers confirmation semantically, verifies only the selected row disappears, preserves the comparison row, checks the expected total and remaining line sum, and verifies the visible badge count.

Static validation passed:

- JSON parsed and TC05-TC08 fixture arithmetic/counts validated.
- `npm run typecheck` passed.
- Playwright `--list` reported exactly TC05-TC08 without executing them.
- Exactly eight FR07 test declarations remain.
- Forbidden-pattern scan passed.

- FR07-TC01-TC04 remain unchanged relative to 0014.
- Scope comparison found no new SUT, FR-01, audit, evidence, database, configuration, documentation, or package changes.

No browser or API tests were run.

## Human review/action

Chấp nhận output 0015. Human action: Em đã review correction và xác nhận FR07-TC05 đến FR07-TC08 không còn dùng exact label “Tổng cộng” hoặc “Tổng tạm tính” để định vị cart total. Summary amount được tìm trong semantic main của cart bằng currency-only regex được neo toàn chuỗi và loại mọi table descendant, nên line amount không thể thỏa locator; exact-label oracle vẫn dành riêng cho FR07-TC10. TC05 giữ đúng một visible Product Detail Add activation cùng các hard fixture prerequisites, không retry hoặc bù hành vi. TC06 giữ red-action và dialog-before-delete oracle; TC07 giữ cancel/no-mutation oracle; TC08 giữ confirmed-delete, total, remaining-row và badge oracle. FR07-TC01 đến FR07-TC04 không bị làm yếu. External JSON có đúng bốn key FR07-TC05 đến FR07-TC08 và không chứa total-label dependency. Typecheck thành công, Playwright chỉ liệt kê đúng 8 FR07 test, forbidden-pattern và whitespace scans đều sạch. Chưa chạy browser/API test và không sửa SUT, FR-01, database, evidence, configuration, documentation hoặc package files.

# Interaction 0016

## AI tool

OpenAI Codex CLI 0.148.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-20T12:26:45Z

## Exact prompt

```text

Complete the reviewed FR-07 Playwright automation with increment 3 covering exactly FR07-TC09 through FR07-TC12.

Read and preserve:

- docs/hw04-traceability-test-matrix.md
- README.md
- playwright.config.ts
- frontend-web/src/App.jsx
- frontend-web/src/context/CartContext.jsx
- frontend-web/src/pages/Home.jsx
- frontend-web/src/pages/ProductDetail.jsx
- frontend-web/src/pages/Cart.jsx
- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/pages/product-detail.page.ts
- tests/support/data-loader.ts

Implement exactly these external JSON records and generated test IDs:

- cart_navigation -> FR07-TC09
- cart_total_formatting -> FR07-TC10
- empty_cart -> FR07-TC11
- product_detail_add_to_cart -> FR07-TC12

Modify only these automation files when necessary:

- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/pages/product-detail.page.ts
- tests/support/data-loader.ts

General constraints:

- The final FR07 spec must contain exactly twelve FR07 test declarations.
- Preserve FR07-TC01 through FR07-TC08 without weakening or removing their assertions.
- Do not modify existing TC01-TC08 test bodies unless a strictly backward-compatible shared-helper adjustment is unavoidable.
- Keep IDs, titles, paths, product fixtures, labels, expected counts, currency rules, initial states, and expected state deltas in external JSON.
- Use only the visible UI for UI fixture construction and target interactions.
- Do not inject localStorage/sessionStorage, access React internals, mock the SUT, compensate for failed clicks, retry target actions, or add arbitrary waits.
- Prefer accessible locators and Playwright web-first assertions.
- Page Object value reads may occur only after suitable web-first existence/visibility/count prerequisites.
- Keep independent requirement observations reachable with soft assertions or guarded reads where one missing feature would otherwise block unrelated evidence.
- Do not invent exact wording, exact URLs, CSS classes, colors, separator characters, or feedback text where the matrix says EVIDENCE MISSING.
- Do not modify SUT source, FR-01 automation, audit files, evidence, database, configuration, documentation, or package files.

FR07-TC09:

- Build the cart_navigation fixture deterministically through the visible web UI.
- Verify the cart page is represented as a child page by a semantic breadcrumb.
- Verify the cart navbar entry exposes a programmatic current-page state, such as aria-current; do not infer current state from an unspecified CSS class.
- Independently assert the required visible “Tiếp tục mua sắm” label from external JSON.
- Because a label mismatch must not prevent the navigation oracle, identify the cart page’s semantic home-destination shopping link separately and activate it exactly once.
- Verify the resulting page is the Home page by externally defined destination identity, not by inventing an exact URL.
- Keep breadcrumb, current-state, required-label, and destination observations independently reachable.

FR07-TC10:

- Build cart_total_formatting deterministically through the visible UI with a displayed amount that requires a thousands separator.
- This is the only increment-3 case that owns the exact total-label oracle.
- Assert the summary label is exactly “Tổng cộng” and that “Tổng tạm tính” is not accepted.
- Assert the displayed total contains đ, contains a thousands separator without inventing its locale-specific character, and numerically equals the sum of displayed line amounts.
- Keep the exact-label assertions and independent currency/arithmetic assertions reachable.
- Do not change the label-independent cart-summary helper used by FR07-TC05 through FR07-TC08; add a separate exact-label helper if needed.

FR07-TC11:

- Use the natural fresh-context empty cart; do not delete or mutate an uncontrolled user cart.
- Assert a clear non-empty empty-cart message using semantic structure and requirements.

rement-derived meaning without inventing exact wording.

- Assert a non-empty icon or illustration through semantic image evidence such as an img with meaningful alt, an accessible role=img, or an equivalently accessible illustration.
- Assert that no populated cart product row is shown.
- Keep message, illustration, and no-row assertions independently reachable.

FR07-TC12:

- Start in a fresh context with the target product absent and the externally defined known badge count.
- Navigate to the target Product Detail through the visible web UI.
- Verify every displayed target product image has a non-empty descriptive alt, including product identity where appropriate.
- Locate the visible cart badge semantically in or associated with the Giỏ hàng navigation entry.
- Missing badge evidence must fail its own assertion but must not block the independent image-alt, single Add activation, and feedback observations.
- Activate the visible Product Detail Add control exactly once. Never retry or compensate if the SUT ignores that activation.
- Assert deterministic visible add-to-cart feedback without inventing exact feedback wording; a visible semantic notification or a visible control-state/text transition is acceptable.
- If a badge is available, verify its numeric value changes from the external initial value by exactly one. Guard numeric reads so a missing badge remains a recorded failure rather than causing unrelated evidence to be skipped.
- Do not claim cart mutation unless it is externally observed.

Page Object expectations:

- Reuse CartPage, CatalogPage, and ProductDetailPage.
- Add robust helpers only where needed for semantic breadcrumb/current navigation state, the exact TC10 total label, empty-state message/illustration, Home destination identity, product image alt, visual feedback, and cart badge lookup/parsing.
- Keep exact-label and label-independent total helpers separate.
- Keep helpers generic and strict TypeScript-compatible.
- Remain compatible with Playwright 1.62.1.

Static validation only:

- Parse and validate the external JSON.
- Run npm run typecheck.
- List the four new tests with Playwright --list without executing them.
- Confirm exactly twelve FR07 declarations.
- Confirm FR07-TC01 through FR07-TC08 remain present and are not weakened.
- Confirm TC12 contains exactly one visible Add activation.
- Scan for forbidden patterns including waitForTimeout, test.skip, test.fixme, retries, duplicate-click helpers, storage injection, and React-internal access.
- Run whitespace checks on authored automation files.
- Confirm no SUT, FR-01, audit, evidence, database, configuration, documentation, or package file changed.

After editing, report:

- exact automation files changed;
 - the four added JSON keys and generated IDs;
 - the TC09 breadcrumb/current-state/required-label/navigation oracles;
 - the TC10 exact-label/currency/arithmetic oracles;
 - the TC11 message/illustration/no-row oracles;
 - the TC12 image-alt/single-Add/feedback/badge-delta oracles and evidence guards;
 - static validation performed;
 - confirmation that no browser/API tests ran and no prohibited file was modified
- ...

Exact AI output

```text

Implemented FR-07 increment 3 with exactly FR07-TC09 through FR07-TC12.

## ### Files changed

- [fr07-cart.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr07-cart.json)
- [fr07-cart.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr07-cart.spec.ts)
- [cart.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/cart.page.ts)
- [catalog.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/catalog.page.ts)
- [product-detail.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/product-detail.page.ts)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)

## ### Added records and generated IDs

- `cart\_navigation` → `FR07-TC09`
- `cart\_total\_formatting` → `FR07-TC10`
- `empty\_cart` → `FR07-TC11`
- `product\_detail\_add\_to\_cart` → `FR07-TC12`

## ### Oracles

- TC09: named semantic breadcrumb containing the current cart page; navbar `aria-current="page"`; independent exact `Tiếp tục mua sắm` assertion; separate meaning-based shopping link activated once; Home identified by its external heading rather than URL.
- TC10: exact `Tổng cộng` label and explicit rejection of `Tổng tạm tính`; `đ`; locale-agnostic thousands-separator detection; displayed total equals the external expected value and sum of displayed line amounts.
- TC11: natural fresh-context cart; meaningful semantic empty message; accessible named illustration; zero populated product rows.
- TC12: every visible target image has non-empty alt containing product identity; guarded semantic badge reads; exactly one visible Product Detail Add activation; semantic notification or control transition feedback; available badge must equal external initial count plus one. Missing badge evidence remains a soft failure and does not block other observations.

## ### Static validation

- Parsed and validated exactly 12 unique JSON records and sequential IDs `FR07-TC01`-`FR07-TC12`.
- `npm run typecheck` passed.
- Playwright `--list` returned exactly the four new Chromium tests.
- Confirmed exactly 12 FR-07 test declarations.
- TC01-TC08 bodies are unchanged; the diff contains additions only.
- Confirmed TC12 has one `addToCartOnce` call, whose helper contains one click.
- Forbidden-pattern scan found no matches.
- `git diff --check` passed.
- Tracked changes are limited to the six allowed automation files.

No browser or API tests were executed. No SUT, FR-01, audit, evidence, database, configuration, documentation, or package file was modified. The three pre-existing untracked `ai-audit` files were left untouched.

## Human review/action

Từ chối output 0016. Human action: Em đã review toàn bộ diff FR-07 increment 3 và phát hiện TC12 chưa sử dụng targetProduct.quantity nên chưa chứng minh Add đúng một đơn vị. visualAddFeedback có thể chấp nhận alert hoặc status tồn tại trước thao tác nên thiếu before/after feedback delta. semanticCartBadge tìm số trong toàn bộ parent navigation nên chưa chứng minh số đó thực sự thuộc hoặc được liên kết với Giỏ hàng. Ngoài ra TC09 và TC12 lặp soft count/visibility trên locator bị thiếu, có nguy cơ hết test budget trước khi hoàn thành các oracle độc lập. Cần lưu proposal hiện tại làm rejected artifact và sửa bằng visible quantity prerequisite, feedback state transition, badge association chặt chẽ, cùng guarded visibility hoặc test.slow phù hợp. Chưa chạy Playwright browser/API test và SUT không bị sửa.

# Interaction 0017

## AI tool

OpenAI Codex CLI 0.148.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-20T13:13:05Z

## Exact prompt

```text

Correct the rejected FR-07 increment 3 proposal for exactly FR07-TC09 through FR07-TC12.

Read and preserve:

- docs/hw04-traceability-test-matrix.md
- README.md
- playwright.config.ts
- frontend-web/src/App.jsx
- frontend-web/src/context/CartContext.jsx
- frontend-web/src/pages/Home.jsx
- frontend-web/src/pages/ProductDetail.jsx
- frontend-web/src/pages/Cart.jsx
- ai-audit/prompts/0016-complete-fr07-cart.txt
- ai-audit/outputs/0016-complete-fr07-cart.md
- ai-audit/entries/0016-complete-fr07-cart.md
- ai-audit/artifacts/0016-rejected-fr07-increment3/fr07-cart.json
- ai-audit/artifacts/0016-rejected-fr07-increment3/fr07-cart.spec.ts
- ai-audit/artifacts/0016-rejected-fr07-increment3/cart.page.ts
- ai-audit/artifacts/0016-rejected-fr07-increment3/catalog.page.ts
- ai-audit/artifacts/0016-rejected-fr07-increment3/product-detail.page.ts
- ai-audit/artifacts/0016-rejected-fr07-increment3/data-loader.ts
- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/pages/product-detail.page.ts
- tests/support/data-loader.ts

The 0016 proposal was rejected for these specific reasons:

1. TC12 declares targetProduct.quantity but never uses it, so it does not prove that exactly one unit is submitted.
2. visualAddFeedback accepts any existing alert or status and lacks a before/after feedback delta.
3. semanticCartBadge searches the whole parent navigation, so an unrelated number

ic element can be mistaken for the cart badge.

4. TC09 and TC12 repeat soft count and visibility waits for missing locators, risking exhaustion of the test budget before independent evidence completes.

Modify only these automation files when necessary:

- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/cart.page.ts
- tests/pages/catalog.page.ts
- tests/pages/product-detail.page.ts
- tests/support/data-loader.ts

Required corrections:

TC12 visible quantity:

- Use targetProduct.quantity from external JSON.
- Scope the quantity control to the displayed target Product Detail.
- Establish a web-first visible quantity-control prerequisite.
- Set or verify the visible quantity as exactly the external value 1, then assert its displayed value before Add.
- Activate the visible Product Detail Add control exactly once.
- Do not retry, compensate, or perform a second Add if the SUT ignores the first activation.

TC12 feedback transition:

- Capture a pre-action feedback baseline before clicking Add.
- A pre-existing unchanged alert, status, or unrelated notification must never satisfy the post-action oracle.
- After the one Add activation, require an externally observable transition:
 - a new or changed semantic alert/status associated with the action; or
 - a change in the scoped Add control's visible text or relevant accessible state such as disabled, aria-disabled, aria-pressed, or aria-busy.
- Compare post-action evidence with the captured baseline.
- Keep feedback wording unspecified because exact wording is EVIDENCE MISSING.
- Missing feedback must remain a failure without preventing independent image-alt or badge evidence.

TC12 cart badge association:

- Do not search all numeric text in the navigation or the unrestricted parent of the cart link.
- A badge may qualify only when it is a numeric descendant of the Giỏ hàng link or is explicitly associated with that link through an accessible relationship such as aria-describedby or aria-labelledby.
- Keep badge parsing strict and guarded.
- Missing initial or updated badge must fail its own soft assertion but must not block quantity, image-alt, Add, or feedback evidence.
- If both badge observations exist, verify the external initial value and exact +1 delta.

TC09 and TC12 evidence budget:

- Avoid unconditional toBeVisible calls after a soft count assertion has already shown that a locator is absent.
- Guard visibility and value reads behind the externally expected count.
- Add test.slow() to TC09 and TC12 because they intentionally collect several independent soft requirement failures.
- Keep breadcrumb, navbar current state, required label, navigation destination, image alt, quantity, feedback, and badge observations independently reachable.

Preservation requirements:

- Keep exactly twelve FR07 test declarations.
- Preserve FR07-TC01 through FR07-TC08 byte-for-byte.
- Preserve the accepted isolation between TC05-TC08 label-independent totals and

TC10 exact-label totals.

- Preserve TC10 and TC11 from the rejected proposal unless a correction is strictly necessary.
- Preserve external JSON keys and IDs:
 - cart_navigation -> FR07-TC09
 - cart_total_formatting -> FR07-TC10
 - empty_cart -> FR07-TC11
 - product_detail_add_to_cart -> FR07-TC12
- Do not weaken any oracle.
- Do not add arbitrary waits, retries, skips, fixmes, storage injection, mocking, React-internal access, or duplicate-click helpers.
- Remain strict TypeScript-compatible and Playwright 1.62.1-compatible.
- Do not modify SUT, FR-01, audit, evidence, database, configuration, documentation, or package files.

Static validation only:

- Parse and validate all 12 external JSON records and sequential IDs.
- Run npm run typecheck.
- Use Playwright --list for TC09-TC12 without executing tests.
- Confirm exactly 12 FR07 declarations.
- Confirm TC01-TC08 are byte-identical to the 0016 rejected artifact.
- Confirm TC10-TC11 remain unchanged unless explicitly justified.
- Confirm TC12 consumes targetProduct.quantity and contains exactly one Add activation.
- Confirm the feedback oracle compares before and after states.
- Confirm the badge locator cannot select unrelated numeric navigation content.
- Scan forbidden patterns and run whitespace checks.
- Confirm no prohibited file changed beyond changes already present before this correction.

After editing, report:

- exact automation files changed relative to the 0016 rejected artifact;
- how TC12 proves visible quantity 1 before exactly one Add;
- how feedback before/after transition is established;
- how the badge is semantically associated with Giỏ hàng;
- how TC09/TC12 avoid evidence-budget exhaustion;
- retained TC09-TC12 oracles;
- static validation performed;
- confirmation that no browser/API test ran and no prohibited file was modified.

Exact AI output

```text

Corrected the rejected FR-07 increment 3 proposal.

Changed relative to the 0016 artifact:

- [fr07-cart.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr07-cart.json)
- [fr07-cart.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr07-cart.spec.ts)
- [cart.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/cart.page.ts)
- [product-detail.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/product-detail.page.ts)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)

`catalog.page.ts` remains byte-identical to the rejected artifact.

Key corrections:

- TC12 scopes the spinbutton to the Product Detail main containing the target product heading. It requires the external `quantityControls: 1`, guards visibility behind that count, and asserts the displayed value equals `String(targetProduct.quantity)`-external value `1`-before Add.
- TC12 calls `addToCartOnce` exactly once; that helper contains exactly one `.click()`. No compensating or repeated Add exists.
- Feedback baseline is captured before Add, including visible target-detail alert/status elements and the exact Add control's text, disabled state, `aria-disabled`, `aria-pressed`, and `aria-busy`.
- Post-action feedback must differ from that baseline through a new/changed scoped semantic signal or a changed state/text on the same Add control. Unchanged pre-existing feedback cannot pass. The soft post-action observation does not prevent badge evidence.
- Badge lookup now accepts only strict numeric descendants of the `Giỏ hàng` link or strict numeric elements referenced by that link through `aria-describedby`/`aria-labelledby`. The unrestricted parent-navigation search was removed.
- Initial and updated badge count/visibility/parsing are guarded. Both available observations prove the external initial value and exact `+1` delta.
- TC09 and TC12 use `test.slow()`. Visibility and value reads are conditional on externally expected locator counts, avoiding repeated waits for absent elements.

#### Retained oracles:

- TC09: semantic breadcrumb, navbar current state, exact required label, independent shopping-destination activation, and Home heading identity.
- TC10: exact `Tổng cộng`, rejection of `Tổng tạm tính`, currency symbol, thousands separator, expected total, and displayed arithmetic.
- TC11: natural empty state, meaningful message, accessible illustration, and zero product rows.
- TC12: image alt, visible quantity `1`, one Add, feedback transition, and associated badge delta.

#### Static validation passed:

- Parsed and validated 12 JSON records with sequential unique IDs `FR07-TC01`-`FR07-TC12`.
- `npm run typecheck`.
- Playwright `--list` returned exactly TC09-TC12 for Chromium.
- Exactly 12 FR07 test declarations.
- TC01-TC08 byte-identical to the rejected artifact.
- TC10-TC11 byte-identical to the rejected artifact.
- Confirmed external quantity consumption, one Add activation, baseline-before-Add ordering, post-action comparison, and restricted badge association.
- Forbidden-pattern scan found no matches.
- `git diff --check` passed.
- Final changed-path set matches the pre-correction baseline; pre-existing audit changes were untouched, and no prohibited file was newly modified.

No browser or API test was executed.

#### ## Human review/action

Từ chối output 0017. Human action: Em đã review correction diff và xác nhận feedback baseline/delta, badge association, test.slow cùng guarded visibility đã được sửa đúng. Tuy nhiên TC12 vẫn dùng soft assertions cho quantity count, visibility và value rồi thực hiện Add vô điều kiện. Vì vậy quantity chưa phải hard fixture prerequisite và post-Add feedback/badge evidence có thể được thu dưới trạng thái không bảo đảm đúng một đơn vị. Cần lưu proposal 0017 làm rejected artifact và sửa quantity thành hard web-first prerequisite hoặc đặt bằng visible control r

ôi hard-assert trước đúng một Add activation. Chưa chạy Playwright browser/API test và SUT không bị sửa.

# Interaction 0018

## AI tool

OpenAI Codex CLI 0.148.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-20T13:41:36Z

## Exact prompt

```text

Apply one narrow final correction to the rejected FR-07 increment 3 proposal.

Read:

- docs/hw04-traceability-test-matrix.md
- ai-audit/entries/0017-correct-fr07-cart-increment3.md
- ai-audit/artifacts/0017-rejected-fr07-increment3/fr07-cart.spec.ts
- tests/data/fr07-cart.json
- tests/features/fr07-cart.spec.ts
- tests/pages/product-detail.page.ts

The 0017 proposal correctly fixed:

- TC09 and TC12 evidence budgets with test.slow and guarded reads;
- TC12 feedback before/after transition;
- TC12 semantic cart-badge association;
- TC12 external quantity field/type declarations;
- preservation of TC01-TC08 and TC10-TC11.

The only remaining defect is that TC12 uses soft assertions for the quantity count, visibility, and value, then performs Add even when the one-unit fixture is invalid.

Modify only:

- tests/features/fr07-cart.spec.ts

Required narrow correction:

- In FR07-TC12, make the target Product Detail quantity control a hard web-first fixture prerequisite before the Add action.
- Require exactly the external expected quantity-control count.
- Require the scoped quantity control to be visible.
- Use the existing visible ProductDetailPage quantity interaction to set targetProduct.quantity exactly once, or retain the current value only if it is hard-asserted.
- Hard-assert the displayed value equals String(testCase.targetProduct.quantity) before Add.
- Only after all quantity prerequisites pass, capture the feedback baseline and activate Add.
- Keep exactly one addToCartOnce call and exactly one click inside that helper.
- Never retry or compensate if the Add activation is ignored.

Preservation:

- Keep FR07-TC01 through FR07-TC11 byte-identical to the 0017 rejected artifact.
- Within TC12, keep image-alt, initial badge, feedback baseline/delta, updated badge, test.slow, guarded reads, and all other assertions unchanged.
- Do not modify JSON, Page Objects, data-loader, SUT, FR-01, audit, evidence, database, configuration, documentation, or package files.
- Do not weaken any oracle.

- Do not add waits, retries, skips, fixmes, storage injection, mocks, React-inte
- rnal access, or duplicate Add actions.
- Keep exactly twelve FR07 test declarations.
- Remain strict TypeScript-compatible and Playwright 1.62.1-compatible.

Static validation only:

- Run npm run typecheck.
- Use Playwright --list for TC09-TC12 without executing tests.
- Confirm the only automation file changed relative to the 0017 rejected artifact is tests/features/fr07-cart.spec.ts.
- Confirm the correction diff is limited to the TC12 quantity prerequisite block
-
- Confirm TC01-TC11 are byte-identical to the 0017 artifact.
- Confirm TC12 consumes targetProduct.quantity through a hard visible prerequisi
- te before exactly one Add.
- Confirm feedback baseline remains before Add.
- Confirm forbidden-pattern and whitespace checks pass.

After editing, report:

- the exact file changed;
- the exact hard quantity sequence;
- confirmation that Add remains exactly once;
- preservation checks;
- static validation performed;
- confirmation that no browser/API test ran and no prohibited file changed.

Exact AI output

```text

Corrected exactly one file: [tests/features/fr07-cart.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr07-cart.spec.ts:1505).

TC12 now hard-requires this sequence before Add:

1. Exact external `quantityControls` count.
2. Scoped quantity control visibility.
3. Displayed value equal to `String(testCase.targetProduct.quantity)`.

The existing value is retained only after that hard assertion. Feedback baseline capture remains after these prerequisites and before Add.

Confirmed:

- Exactly one TC12 `addToCartOnce` call.
- Its unchanged helper contains exactly one click.
- No retry or compensating Add.
- Exactly 12 FR07 declarations.
- TC01-TC11 are byte-identical to the 0017 artifact.
- The diff is confined to TC12's quantity prerequisite block.
- All other 0017 automation files are byte-identical to their artifacts.
- Forbidden-pattern scan and `git diff --check` passed.
- `npm run typecheck` passed.
- Playwright `--list` found exactly TC09-TC12 for Chromium.

No browser/API test ran, and no prohibited file was changed.

## Human review/action

Chấp nhận output 0018. Human action: Em đã review final correction và xác nhận c  
hỉ tests/features/fr07-cart.spec.ts thay đổi so với rejected artifact 0017, tại

đúng quantity prerequisite block của FR07-TC12. Quantity control được yêu cầu đúng external count, visible và có displayed value bằng String(targetProduct.quantity) dưới hard web-first assertions trước feedback baseline và đúng một addToCartOnce activation. Feedback before/after transition, semantic badge association, guarded reads, test.slow, image-alt và badge-delta oracles được giữ nguyên. Năm automation file còn lại byte-identical với artifact 0017; FR07-TC01 đến FR07-TC11 không thay đổi. Spec vẫn có đúng 12 FR07 declarations, typecheck thành công, Playwright chỉ liệt kê đúng TC09-TC12, forbidden-pattern và whitespace checks đều sạch. Chưa chạy Playwright browser/API test và không sửa SUT, FR-01, database, evidence, configuration, documentation hoặc package files.

# Interaction 0019

## AI tool

OpenAI Codex CLI 0.148.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-21T02:21:03Z

## Exact prompt

```text

Implement the first reviewed FR-12 Playwright automation increment covering exactly FR12-TC01 through FR12-TC04.

Read and preserve:

- README.md
- api_specification.md
- docs/hw04-traceability-test-matrix.md
- playwright.config.ts
- frontend-admin/vite.config.js
- frontend-admin/src/App.jsx
- frontend-web/src/App.jsx
- frontend-web/src/pages/Login.jsx
- frontend-web/src/context/AuthContext.jsx
- backend/database.js
- backend/server.js
- tests/data/fr01-registration.json
- tests/data/fr07-cart.json
- tests/features/fr01-registration.spec.ts
- tests/features/fr07-cart.spec.ts
- tests/pages/registration.page.ts
- tests/support/data-loader.ts

Implement exactly these external JSON records and generated test IDs:

- admin_ui_no_token -> FR12-TC01
- admin_ui_ordinary_user -> FR12-TC02
- admin_ui_admin -> FR12-TC03
- admin_route_inventory_missing_token -> FR12-TC04

Create or modify only:

- tests/data/fr12-access-control.json
- tests/features/fr12-access-control.spec.ts
- tests/pages/admin-access.page.ts
- tests/support/fr12-api.ts
- tests/support/data-loader.ts

Do not modify:

- SUT source;

- FR01 or FR07 data, specs, Page Objects, or helpers;
- audit files;
- evidence;
- database;
- Playwright configuration;
- documentation;
- package files.

General requirements:

- Add exactly four FR12 test declarations.
- Keep all paths, credentials, labels, expected counts, route inventory, HTTP methods, safe IDs, payloads, mutation markers, and expected authorization partitions in external JSON.
- Use PW_ADMIN_URL with fallback http://localhost:5174 for the admin origin; keep the admin entry path `/` in external JSON.
- Use PW_API_URL with fallback http://localhost:3000 for API requests.
- Do not inject localStorage or sessionStorage.
- Do not access React internals, mock the SUT, retry target actions, compensate for ignored actions, use arbitrary waits, or alter the runtime database directly.
- Prefer accessible locators and Playwright web-first assertions.
- Do not invent exact denial wording, exact denial status codes, or exact response bodies where the matrix says EVIDENCE MISSING.
- Distinguish authentication from authorization.
- Keep independent observations reachable with soft assertions and guarded reads where appropriate.
- Remain strict TypeScript-compatible and Playwright 1.62.1-compatible.

FR12-TC01:

- Start with a natural fresh browser context and visit the external admin entry path directly.
- Do not read or write browser storage.
- Assert that the visible Admin login form is available using externally defined identity and controls.
- Assert that protected admin identity/content is not visible.
- Observe requests during the direct visit and assert that no admin mutation request is issued.
- Treat this as authentication coverage.

FR12-TC02:

- Use the externally defined seeded ordinary-user credentials.
- Visit the Admin login UI and submit the visible login form exactly once.
- Do not obtain or inject a token outside the UI.
- Require deterministic evidence that the ordinary user is denied Admin UI access without inventing denial wording. An accessible browser dialog, remaining on the login form, and absence of protected admin identity are acceptable independent observations.
- Handle any browser dialog deterministically.
- Assert no admin mutation request is issued.
- Treat this as authorization coverage: the credentials are valid, but the role is not admin.

FR12-TC03:

- Use the externally defined seeded admin credentials.
- Visit the Admin login UI and submit the visible login form exactly once.
- Do not obtain or inject a token outside the UI.
- Assert that the protected Admin identity and externally defined Dashboard identity become visible.
- Assert the login form is no longer visible.
- Keep the success identity observations independent.
- Perform no admin mutation.

FR12-TC04:

- Exercise every registered and documented ``/api/admin/*`` route from external JS ON with no Authorization header:
 - POST `/api/admin/import-products`
 - POST `/api/admin/coupons`
 - DELETE `/api/admin/coupons/:id`
 - GET `/api/admin/users`
 - DELETE `/api/admin/users/:id`
 - GET `/api/admin/orders`
 - PUT `/api/admin/orders/:id/status`
- Use non-destructive denial probes.
- Mutation-shaped POST probes must use valid-looking externally defined payloads with unique deterministic markers so reaching mutation code would be externally detectable.
- DELETE and PUT probes must use externally defined non-existing safe IDs.
- Do not assert exact 401/403 values because the matrix marks exact status as EV IDENCE MISSING. Assert a non-success response and that protected collections/data are not returned.
- Establish controlled pre/post state snapshots through supported read APIs using an independently acquired seeded-admin API session only for snapshot setup and verification. The target seven route probes themselves must contain no Authorization header.
- Snapshot representative users, orders, products, and coupons before the probes and compare them after all probes.
- Guard snapshot-dependent assertions so a snapshot acquisition failure is recorded without hiding independent denial results for the seven routes.
- Assert each route result independently.
- Assert that no controlled mutation marker appears and no protected record changes.
- Do not mutate or restore the SQLite file directly.
- Treat this as authentication coverage.

Page Object and helper requirements:

- AdminAccessPage must scope the login form, email control, password control, submit button, protected Admin identity, and Dashboard identity.
- The login helper must perform exactly one visible submit activation.
- Request-observation helpers must classify mutation requests by external method/path data.
- FR12 API helpers must build requests from external route records, omit Authorization for target missing-token probes, parse responses defensively, and compare normalized snapshots.
- Do not place inline test-case arrays or route inventories in the spec.

Static validation only:

- Parse and validate the FR12 JSON.
- Confirm exactly four unique sequential IDs FR12-TC01 through FR12-TC04.
- Run `npm run typecheck`.
- Use Playwright `--list` to list only FR12-TC01 through FR12-TC04 without executing them.
- Confirm exactly four FR12 declarations.
- Confirm FR01 and FR07 tracked files remain byte-identical to HEAD.
- Confirm TC02 and TC03 each contain exactly one visible login submit activation.
- Confirm all seven TC04 target probes omit Authorization.
- Confirm TC04 contains all seven required admin routes exactly once.
- Scan for `waitForTimeout`, `test.skip`, `test.fixme`, retries, storage injection, mocks, React-internal access, duplicate-submit helpers, and direct SQLite access.
- Run whitespace checks.
- Confirm no prohibited file changed.

After editing, report:

- exact files created or changed;

- four JSON keys and IDs;
- TC01 authentication oracles;
- TC02 ordinary-user authorization oracles;
- TC03 positive admin authorization oracles;
- TC04 seven-route denial and no-mutation design;
- evidence guards;
- static validation performed;
- confirmation that no browser/API test ran and no prohibited file changed.

Exact AI output

```text

Implemented the first FR-12 increment with exactly four declarations.

### Files:

- Created [fr12-access-control.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr12-access-control.json)
- Created [fr12-access-control.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr12-access-control.spec.ts:53)
- Created [admin-access.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/admin-access.page.ts:56)
- Created [fr12-api.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/fr12-api.ts:128)
- Modified [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts:657)

### JSON mappings:

- `admin\_ui\_no\_token` → `FR12-TC01`
- `admin\_ui\_ordinary\_user` → `FR12-TC02`
- `admin\_ui\_admin` → `FR12-TC03`
- `admin\_route\_inventory\_missing\_token` → `FR12-TC04`

### Coverage:

- TC01 authentication: natural fresh context, direct `/` visit, visible scoped login form and controls, protected Admin/Dashboard identities absent, no login or admin mutation request.
- TC02 authorization: exactly one visible login submission, successful ordinary-user login response observed without reading its body/token, dialog deterministically dismissed without asserting wording, login form remains, protected identities remain absent, no admin mutation.
- TC03 positive authorization: exactly one visible login submission, successful login response, independent Admin and Dashboard visibility assertions, login form and controls removed, no mutation.
- TC04 authentication: all seven required routes are probed once without `Authorization`; POST probes use marked valid-looking payloads, while DELETE/PUT use non-existing safe IDs. Each response is independently checked for non-success and absence of protected data without asserting exact status or body.

TC04 captures users, orders, products, and coupons through a separately acquired seeded-admin snapshot session. Pre/post snapshots are normalized before comparison, markers are scanned across controlled state, and snapshot failures are recorded with soft assertions while all seven denial probes remain reachable.

### Static validation passed:

- JSON parsing, exact keys, sequential unique IDs, route inventory, markers, safe IDs, and null authorization configuration
- `npm run typecheck`

- FR12-only Playwright `--list`: four declarations across three configured projects
- Exactly one TC02 and one TC03 login-helper call; helper contains one click
- FR01/FR07 tracked files byte-identical to `HEAD`
- Forbidden-pattern and whitespace scans
- `git diff --check`
- Required/reference and prohibited files unchanged

No browser or API test was executed—only TypeScript checking and Playwright discovery. No database, SUT, configuration, documentation, evidence, package, or audit file was modified; the two pre-existing untracked audit files were left untouched.

## Human review/action

Từ chối output 0019. Human action: Em đã review toàn bộ FR-12 increment 1 diff và xác nhận bốn external JSON key/ID, inventory đủ bảy /api/admin/\* routes, target probes không truyền Authorization, snapshot session tách biệt, và mỗi TC02/TC03 chỉ có một loginOnce activation. Tuy nhiên TC02 và TC03 dùng soft login-form/control prerequisites rồi thực hiện loginOnce vô điều kiện, nên authorization evidence có thể được thu dưới fixture UI không hợp lệ hoặc hết test budget. TC01 đến TC03 cũng lập soft count/visibility trên locator bị thiếu mà không guard. Ngoài ra TC04 chỉ phát hiện unexpected mutation qua snapshot/marker nhưng không cleanup product/coupon sentinel nếu SUT cho phép mutation, có thể làm bẩn runtime database. Cần lưu proposal hiện tại làm rejected artifact, sửa TC02/TC03 thành hard visible login prerequisites, guard các independent UI observations hoặc dùng test.slow phù hợp, và cleanup controlled mutations qua API trong finally mà không sửa SQLite trực tiếp. Chưa chạy Playwright browser/API test và SUT không bị sửa.

# Interaction 0020

## AI tool

OpenAI Codex CLI 0.148.0 (model: gpt-5.6-sol)

## ISO date/time

2026-08-21T02:52:09Z

## Exact prompt

```text

Correct the rejected FR-12 increment 1 proposal covering exactly FR12-TC01 through FR12-TC04.

Read and preserve:

- README.md
- api_specification.md
- docs/hw04-traceability-test-matrix.md
- playwright.config.ts
- frontend-admin/vite.config.js
- frontend-admin/src/App.jsx
- backend/database.js
- backend/server.js
- ai-audit/prompts/0019-add-fr12-access-control-increment1.txt
- ai-audit/outputs/0019-add-fr12-access-control-increment1.md
- ai-audit/entries/0019-add-fr12-access-control-increment1.md
- ai-audit/artifacts/0019-rejected-fr12-increment1/fr12-access-control.json
- ai-audit/artifacts/0019-rejected-fr12-increment1/fr12-access-control.spec.ts
- ai-audit/artifacts/0019-rejected-fr12-increment1/admin-access.page.ts

- ai-audit/artifacts/0019-rejected-fr12-increment1/fr12-api.ts
- ai-audit/artifacts/0019-rejected-fr12-increment1/data-loader.ts
- tests/data/fr12-access-control.json
- tests/features/fr12-access-control.spec.ts
- tests/pages/admin-access.page.ts
- tests/support/fr12-api.ts
- tests/support/data-loader.ts

The 0019 proposal was rejected for these reasons:

1. TC02 and TC03 use soft login-form/control prerequisites and then call loginOnce unconditionally, so authorization evidence can be collected under an invalid UI fixture or the test can exhaust its timeout.
2. TC01 through TC03 repeat soft count and visibility waits on missing locators without guards.
3. TC04 detects unexpected mutation but does not clean controlled product/coupon sentinels through the API, so a failing access-control implementation can contaminate runtime state.

Modify only when necessary:

- tests/data/fr12-access-control.json
- tests/features/fr12-access-control.spec.ts
- tests/support/fr12-api.ts
- tests/support/data-loader.ts

Keep tests/pages/admin-access.page.ts byte-identical to the 0019 rejected artifact unless an unavoidable backward-compatible correction is required and explicitly justified.

Required UI corrections:

TC01:

- Keep the natural fresh context, direct Admin entry visit, login-boundary observations, protected-content absence, and no-mutation observation.
- Add test.slow() because independent soft evidence is intentionally collected.
- Do not perform unconditional toBeVisible after a soft count assertion has shown that a locator is absent.
- Guard visibility observations behind the externally expected count using immediate guarded reads or an equivalent evidence-budget-safe approach.

TC02 and TC03:

- Add test.slow().
- Before loginOnce, make the externally expected login form, email control, password control, and submit control counts hard web-first fixture prerequisites.
- Hard-require each scoped control to be visible.
- Only after all hard prerequisites pass, call loginOnce exactly once.
- Do not retry or compensate if the single login submission is ignored.
- Keep post-action denial/success observations soft and independently reachable.
- Guard post-action visibility reads behind observed expected counts; do not repeat full web-first waits for already missing locators.
- Preserve TC02 valid-login-response, deterministic dialog handling, login-boundary persistence, protected-content absence, and no-mutation oracles.
- Preserve TC03 successful-login response, protected Admin identity, Dashboard identity, login-form disappearance, and no-mutation oracles.

Required TC04 cleanup correction:

- Preserve the exact seven target route probes, their order-independent denial assertions, missing Authorization configuration, payload markers, safe IDs, protected-data checks, and pre/post snapshot defect evidence.
- Keep every target probe free of Authorization.
- Add controlled cleanup configuration to external JSON for only the two possible POST sentinels:
 - imported product identified by exact external marker field/value and removed

- through an external cleanup method/path template;
- coupon identified by exact external marker field/value and removed through a n external cleanup method/path template.
- Cleanup discovery must use successful post-probe snapshots and exact marker-field equality, not substring matching across arbitrary records.
- Extract only strict externally configured identifier fields from matching controlled records.
- Perform cleanup with the separately acquired seeded-admin snapshot session.
- Run cleanup in a finally block after probes so it is attempted even when an assertion/helper unexpectedly throws.
- Never delete an uncontrolled record.
- Never modify SQLite directly.
- Record the original pre-cleanup mutation evidence before cleanup; cleanup must not erase or weaken the defect oracle.
- Capture a final post-cleanup snapshot and assert that controlled markers are absent and controlled state is restored to the pre-probe snapshot where snapshots are available.
- If cleanup or final snapshot acquisition fails, record a failure with diagnostics.
- If no controlled marker exists, cleanup performs zero mutations and final verification still runs.
- Keep users/orders safe-ID probes non-mutating; do not invent cleanup mutations for records that were never controlled.

External-data requirements:

- Preserve exactly these keys and IDs:
 - admin_ui_no_token -> FR12-TC01
 - admin_ui_ordinary_user -> FR12-TC02
 - admin_ui_admin -> FR12-TC03
 - admin_route_inventory_missing_token -> FR12-TC04
- Keep paths, credentials, labels, counts, methods, route inventory, payloads, safe IDs, marker fields, identifier fields, cleanup methods/paths, and expected cleanup outcomes in JSON.
- Keep strict TypeScript interfaces in data-loader.
- Do not add inline test-case arrays or cleanup route objects to the spec.

Preservation:

- Keep exactly four FR12 declarations.
- Preserve all accepted 0019 assertions and helpers unless this correction strictly requires adjustment.
- Preserve FR01 and FR07 byte-for-byte.
- Do not modify SUT, audit, evidence, database, Playwright configuration, documentation, or package files.
- Do not add waits, retries, skips, fixmes, storage injection, mocks, React-internal access, duplicate login actions, or direct database access.
- Remain compatible with strict TypeScript and Playwright 1.62.1.

Static validation only:

- Parse and validate FR12 JSON, four unique sequential IDs, seven unique target routes, two unique controlled cleanup definitions, exact marker fields, and safe identifiers.
- Run npm run typecheck.
- Use Playwright --list for FR12-TC01 through FR12-TC04 without executing tests.
- Confirm exactly four FR12 declarations.
- Confirm TC02 and TC03 hard visible prerequisites occur before exactly one loginOnce call.
- Confirm TC01 through TC03 use test.slow and guarded post-count visibility observations.
- Confirm all seven target probes still omit Authorization.
- Confirm cleanup uses exact controlled marker fields, authenticated API cleanup, and a finally block.
- Confirm original post-probe mutation evidence occurs before cleanup and final

restoration evidence occurs after cleanup.

- Confirm admin-access.page.ts is byte-identical to the 0019 artifact unless explicitly justified.
- Confirm FR01 and FR07 remain byte-identical to HEAD.
- Scan forbidden patterns and run whitespace checks.
- Confirm no prohibited file changed beyond pre-existing audit changes.

After editing, report:

- exact automation files changed relative to the 0019 rejected artifact;
- hard TC02/TC03 login prerequisite sequence;
- guarded evidence-budget correction;
- TC04 controlled cleanup and finally behavior;
- retained TC01-TC04 oracles;
- static validation performed;
- confirmation that no browser/API test ran and no prohibited file was modified.

Exact AI output

```text

Corrected the rejected FR-12 increment 1 proposal for exactly FR12-TC01-FR12-TC04.

### Changed relative to 0019

- [fr12-access-control.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr12-access-control.json:96)
- [fr12-access-control.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr12-access-control.spec.ts:60)
- [fr12-api.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/fr12-api.ts:165)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts:737)

[admin-access.page.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/pages/admin-access.page.ts) remains byte-identical to the rejected artifact; no correction was necessary.

### Corrections

- TC01-TC03 now use `test.slow()`.
- TC02/TC03 hard prerequisite sequence is: form count → form visible → email count → email visible → password count → password visible → submit count → submit visible → exactly one `loginOnce`.
- Soft count/visibility evidence now uses a web-first count followed by an immediate count read; visibility is checked only when the expected positive count is present.
- TC04 retains all seven no-Authorization probes and records original post-probe mutation evidence before cleanup.
- Exact `name` and `code` marker equality discovers controlled product/coupon records. Only strict positive-integer configured `id` fields are deleted through the authenticated seeded-admin session.
- Cleanup runs in nested `finally` teardown. Zero markers cause zero cleanup mutations, while final snapshot and restoration verification still run.
- Cleanup/discovery/final-snapshot failures produce diagnostic soft failures.

All original TC01-TC04 login-boundary, authorization, dialog, protected-content, response, no-mutation, snapshot, marker, and identity oracles remain.

### Static validation

Passed:

- FR12 JSON parsing, four sequential unique IDs, seven unique routes, safe IDs, two cleanup definitions, and exact marker fields
- `npm run typecheck`
- FR12-only Playwright `--list`: four declarations across Chromium, Firefox, and WebKit
- Login prerequisite ordering and single submissions
- Guarded visibility and `test.slow()` checks
- No Authorization in all target probes
- Exact-marker authenticated cleanup, `finally` teardown, and evidence ordering
- FR01/FR07 byte equality with `HEAD`
- Page Object equality with 0019
- Forbidden-pattern, trailing-whitespace, and `git diff --check` scans
- Required SUT, documentation, configuration, database, and audit paths preserved

No browser or API test was executed. The pre-existing audit changes were untouched, and no prohibited file was modified.

## Human review/action

Chấp nhận output 0020. Human action: Em đã review toàn bộ correction diff FR-12 increment 1 và xác nhận TC02/TC03 hard-require đúng external login form/control counts cùng visibility trước đúng một loginOnce activation; không có retry hoặc compensating submit. TC01 đến TC03 dùng test.slow và guarded post-count visibility để bảo toàn evidence budget. TC04 giữ nguyên đủ bảy missing-token probes không truyền Authorization, ghi nhận post-probe snapshot/marker defect evidence trước cleanup, rồi chỉ tìm product/coupon sentinel bằng exact marker-field equality và strict positive-integer ID. Controlled cleanup dùng seeded-admin API session trong nested finally; zero marker không tạo cleanup mutation, còn final snapshot kiểm tra marker vắng mặt và state trở lại pre-probe baseline. Admin Page Object byte-identical với rejected artifact 0019; FR01/FR07 không thay đổi. External JSON có đúng bốn sequential IDs, bảy unique target routes và hai controlled cleanup definitions. Typecheck thành công, Playwright chỉ liệt kê đúng bốn TC01-TC04 trên Chromium, declaration count là 4, forbidden-pattern, whitespace và diff checks đều sạch. Chưa chạy Playwright browser/API test và không sửa SUT, database, evidence, configuration, documentation hoặc package files.

# Interaction 0021

## AI tool

Codex CLI 0.148.0

## ISO date/time

2026-08-21T03:28:59Z

## Exact prompt

```text

Implement the second reviewed FR-12 Playwright automation increment covering exactly FR12-TC05 through FR12-TC08.

Read and preserve:

- README.md
- api_specification.md
- docs/hw04-traceability-test-matrix.md
- playwright.config.ts
- backend/database.js
- backend/server.js

- backend/package.json
- tests/data/fr12-access-control.json
- tests/features/fr12-access-control.spec.ts
- tests/pages/admin-access.page.ts
- tests/support/fr12-api.ts
- tests/support/data-loader.ts
- evidence/development/fr12-increment1-chromium/run-record.md
- ai-audit/entries/0020-correct-fr12-access-control-increment1.md

Implement exactly these additional external JSON records and generated IDs:

- admin_route_inventory_invalid_tokens -> FR12-TC05
- admin_route_inventory_user_token -> FR12-TC06
- admin_read_routes_admin_token -> FR12-TC07
- product_mutation_access_matrix -> FR12-TC08

Modify only when necessary:

- tests/data/fr12-access-control.json
- tests/features/fr12-access-control.spec.ts
- tests/support/fr12-api.ts
- tests/support/data-loader.ts

Keep tests/pages/admin-access.page.ts byte-identical to HEAD.

General constraints:

- The final FR12 spec must contain exactly eight FR12 test declarations.
- Preserve FR12-TC01 through FR12-TC04 test bodies byte-for-byte.
- Preserve all accepted TC01-TC04 helpers and oracles.
- Preserve FR01 and FR07 byte-for-byte.
- Keep all cases, IDs, credentials, token partitions, JWT fixture material, claims, paths, methods, payloads, safe IDs, markers, snapshot resources, expected counts, expected state deltas, cleanup marker fields, cleanup routes, and decision table rows in external JSON.
- API origins use PW_API_URL with fallback http://localhost:3000.
- Do not modify SUT, audit, evidence, database, Playwright configuration, documentation, or package files.
- Do not access SQLite directly.
- Do not inject browser storage, mock the SUT, access React internals, retry target requests, compensate for failed target actions, or add arbitrary waits.
- Do not assert exact denial status codes or response wording where the matrix says EVIDENCE MISSING.
- Keep authentication and authorization observations distinct.
- Prefer independent soft requirement observations after hard fixture prerequisites.
- Use test.slow() where many independent route or matrix observations intentionally run.
- Remain strict TypeScript-compatible and Playwright 1.62.1-compatible.
- Use only Node built-in modules already available to the test project; do not add packages.

FR12-TC05 – invalid-token route inventory:

- Cover every registered/documented /api/admin/* route for exactly two external token classes: malformed and expired.
- Materialize exactly fourteen external probe rows: seven routes for each token class. Do not construct an inline route matrix in the spec.
- The malformed token value must be external.
- Build the expired token deterministically with Node built-in crypto from external JWT header, claims, and source-observed test signing material.
- Hard-validate before probing that the expired fixture:
 - has exactly three JWT segments;
 - uses the external algorithm;
 - has a numeric exp claim earlier than the current time;
 - has a valid signature for the external test signing material.

- This fixture exercises expiry rejection by the observed verifier; do not claim that the SUT normally issues exp-bearing tokens or invent a production expiry contract.
- Send each target probe exactly once with its assigned invalid Bearer token.
- Assert independently that every response is non-success and does not return protected data, without asserting exact status/body.
- Use valid-looking external POST payloads, external non-existing safe IDs, exact token-class-specific mutation markers, pre/post snapshots, and controlled API cleanup in finally.
- Preserve original mutation evidence before cleanup and verify final restoration.
- There must be four controlled POST cleanup markers: import-product and coupon markers for each token class.

FR12-TC06 – ordinary-user token route inventory:

- Acquire the external seeded ordinary-user token through exactly one supported API login.
- Hard-require successful credential authentication and a non-empty token before target probes.
- Establish external role evidence that this fixture is an ordinary user without inventing denial wording.
- Probe each of the same seven Admin routes exactly once with the ordinary-user Bearer token.
- Use valid-looking external POST payloads with exact controlled markers and non-existing safe IDs for mutation-shaped DELETE/PUT routes.
- Assert each route is denied as authorization, protected data is not returned, and controlled state does not change.
- Record all independent route results even if some fail.
- Capture pre/post snapshots using a separate seeded-admin snapshot session.
- Record mutation evidence before cleanup.
- Cleanup only exact controlled product/coupon sentinels via authenticated API in finally, then verify final restoration.
- Never delete an uncontrolled record.

FR12-TC07 – positive admin read authorization:

- Acquire the external seeded-admin token through exactly one supported API login.
- Hard-require successful login, a non-empty token, and external admin-role evidence.
- Exercise only the externally listed read-only Admin routes:
 - GET /api/admin/users
 - GET /api/admin/orders
- Send each read exactly once with the admin Bearer token.
- Assert each request succeeds and returns a protected collection representation without inventing exact status code or complete schema.
- Cross-check externally known seeded user identities where configured.
- Permit an externally configured empty orders collection; do not invent order fixtures.
- Perform no POST, PUT, or DELETE target request and no cleanup mutation.

FR12-TC08 – product mutation access matrix:

- Implement exactly nine external decision-table rows: POST, PUT, and DELETE crossed with missing token, valid ordinary-user token, and valid admin token.
- Acquire ordinary-user and admin tokens through one supported API login each.
- Hard-require both login fixtures, non-empty tokens, and their external role partitions before matrix target actions.
- Hard-require the external category reference through a supported read API before product setup.
- Build all controlled product setup through supported APIs using the admin token. Do not use SQLite.
- Use separate externally defined controlled markers/payloads per decision row so rows cannot mask each other.

- Execute every target matrix row exactly once; no retry or compensation.
- Missing-token rows must be evaluated as authentication denial.
- Ordinary-user rows must be evaluated as authorization denial.
- Admin rows must be evaluated as positive access that reaches business execution.
- For denied POST rows, assert no controlled product appears.
- For denied PUT rows, assert the controlled target remains unchanged.
- For denied DELETE rows, assert the controlled target remains present and unchanged.
- For admin POST, assert the controlled product appears with the external expected fields.
- For admin PUT, assert the controlled target changes to the external expected fields.
- For admin DELETE, assert the controlled target is removed.
- Keep response acceptance/denial and state-delta observations independently reachable where fixture prerequisites exist.
- Record the pre-cleanup matrix state before teardown.
- In finally, discover only exact controlled invariant markers, require strict positive-integer IDs, delete only those controlled products using the configured authenticated cleanup route, and verify the final product collection equals the pre-test baseline.
- Cleanup must also handle unexpected products created or changed by denied rows.
- Never delete or overwrite an uncontrolled product.

JWT/API helper requirements:

- Implement URL-safe base64 and HS256 signing/verification with Node built-in crypto only.
- Keep the signing material and token claims external; never read backend source at runtime.
- Do not log raw tokens or signing material in diagnostics or evidence.
- Generalize authenticated API sessions and target probes without weakening TC04.
- Every target request must make its Authorization partition explicit: absent, invalid Bearer, ordinary-user Bearer, or admin Bearer.
- Snapshot and cleanup helpers must catch transport failures, return guarded diagnostics, and avoid false no-change/no-marker passes when snapshots are unavailable.
- Cleanup discovery must use exact configured field equality and strict IDs.
- Keep all route/matrix collections external, not inline in the spec.

Static validation only:

- Parse and validate all eight FR12 JSON records and sequential unique IDs FR12-TC01 through FR12-TC08.
- Validate TC05 has two token classes, fourteen probe rows, seven unique routes per class, and four cleanup markers.
- Validate TC06 has seven unique route probes and two cleanup markers.
- Validate TC07 has exactly two GET-only routes and no mutation configuration.
- Validate TC08 has exactly nine unique method x token-partition rows and unique controlled markers.
- Run npm run typecheck.
- Use Playwright --list for FR12-TC05 through FR12-TC08 without executing tests.
- Confirm exactly eight FR12 declarations.
- Confirm TC01-TC04 bodies are byte-identical to HEAD.
- Confirm admin-access.page.ts, FR01, and FR07 are byte-identical to HEAD.
- Confirm each login acquisition and each target action occurs exactly once.
- Confirm invalid JWT fixture validation occurs before TC05 probes.
- Confirm TC05/TC06 original mutation evidence precedes finally cleanup.
- Confirm TC08 setup, nine one-shot target actions, pre-cleanup evidence, finally cleanup, and final restoration ordering.
- Confirm no raw token/signing-material logging.
- Scan forbidden patterns and run whitespace checks.

- Confirm no prohibited file changed.

After editing, report:

- exact automation files changed;
- four added JSON keys and IDs;
- TC05 malformed/expired fixture proof and 14-route denial design;
- TC06 ordinary-user seven-route authorization design and cleanup;
- TC07 two positive admin reads;
- TC08 nine-row decision matrix, state deltas, and cleanup;
- evidence guards;
- static validation performed;
- confirmation that no browser/API test ran and no prohibited file changed.

Exact AI output

```text

Implemented the reviewed FR-12 increment for exactly FR12-TC05 through FR12-TC08.

### Changed files

- [fr12-access-control.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr12-access-control.json:358)
- [fr12-access-control.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr12-access-control.spec.ts:802)
- [fr12-api.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/fr12-api.ts:273)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts:868)

Added external records:

- `admin\_route\_inventory\_invalid\_tokens` → `FR12-TC05`
- `admin\_route\_inventory\_user\_token` → `FR12-TC06`
- `admin\_read\_routes\_admin\_token` → `FR12-TC07`
- `product\_mutation\_access\_matrix` → `FR12-TC08`

### Coverage

- TC05 contains exactly fourteen external probes: all seven Admin routes for malformed and expired Bearer tokens. The deterministic HS256 fixture was statically proven to have three segments, matching algorithm, numeric past `exp`, and a valid signature. Four exact product/coupon cleanup markers restore state without asserting a production expiry contract.
- TC06 performs one ordinary-user login, validates external `user` role evidence, probes all seven Admin routes once, records independent authorization failures and state evidence, then cleans only the two exact sentinels through the seeded -admin API session.
- TC07 performs one admin login and exactly two GET target requests for `/api/admin/users` and `/api/admin/orders`. It verifies collection representations, configured seeded user identities, and permits empty orders.
- TC08 implements exactly nine unique POST/PUT/DELETE × missing/user/admin rows. All setup uses supported APIs with the admin token and a read API category reference. Each target runs once with independent response/state assertions. Final cleanup discovers exact invariant names, accepts only positive integer IDs, removes only controlled products, and verifies the complete pre-test baseline.

Transport, snapshot, cleanup, and final-restoration failures produce guarded diagnostics and cannot become false no-change/no-marker passes. Tokens and signing material are never logged.

### ### Static validation

Passed:

- JSON parsing and sequential unique `FR12-TC01`-`FR12-TC08` IDs
- TC05: 2 token classes, 14 probes, 7 routes/class, 4 cleanup markers
- TC06: 7 routes and 2 cleanup markers
- TC07: exactly 2 GET-only routes with no mutation/cleanup configuration
- TC08: 9 unique decisions, 9 unique primary markers, 12 unique cleanup invariants
- `npm run typecheck`
- Playwright `--list` for TC05-TC08: four cases across Chromium, Firefox, and WebKit
- Exactly eight FR12 declarations overall
- TC01-TC04 describe block byte-identical to HEAD
- `admin-access.page.ts`, FR01, and FR07 byte-identical to HEAD
- One-shot login/target structure and required evidence/cleanup ordering
- Forbidden-pattern, raw-secret logging, trailing-whitespace, and `git diff --check` scans

No browser or API test was executed. Only the four permitted automation files changed; SUT, configuration, documentation, database, package, audit, evidence, and Admin Page Object files were not modified.

### ## Human review/action

Từ chối output 0021. Human action: Em đã review toàn bộ FR-12 increment 2 diff và xác nhận external JSON bổ sung đúng FR12-TC05 đến FR12-TC08; TC05 có đủ 14 malformed/expired-token probes trên bảy Admin routes, TC06 có đủ bảy ordinary-user probes, TC07 chỉ thực hiện hai admin GET reads, và TC08 có đúng chín POST/PUT/DELETE x missing/user/admin target actions với controlled setup, exact-marker cleanup và final baseline verification. Tuy nhiên TC05 và TC06 probe PUT /api/admin/orders/:id/status bằng safe non-existing IDs và chỉ yêu cầu non-success response cùng unchanged snapshots. Nếu authentication/authorization bị bypass, business handler vẫn có thể trả 404 Order not found mà không đổi state; các oracle hiện tại sẽ hiểu nhầm kết quả này là access denial. Cần lưu proposal 0021 làm rejected artifact và sửa order-status probes dùng controlled existing order có business-valid transition, kiểm tra trạng thái không đổi dưới invalid/ordinary-user token, rồi restore controlled state qua authenticated admin API trong finally. Nếu không thể thiết lập fixture hợp lệ thì phải ghi rõ evidence not reached, không được phân loại mọi non-success response là authentication/authorization denial. Chưa chạy Playwright browser/API test và không sửa SUT.

### # Interaction 0022

#### ## AI tool

Codex CLI 0.148.0

#### ## ISO date/time

2026-08-21T03:57:10Z

#### ## Exact prompt

```text

Act as a senior Playwright TypeScript test-automation engineer correcting the rejected FR-12 increment 2 proposal for HW04 student 23127027.

Repository root:

`/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027`

Read before editing:

- `README.md`, especially FR-12, SEC-02, SEC-03, and relevant FR-21 through FR-24 requirements.
- `docs/hw04-traceability-test-matrix.md`, especially FR12-TC05 through FR12-TC08.
- `api\_specification.md` only as supporting contract.
- Current:
 - `tests/data/fr12-access-control.json`
 - `tests/features/fr12-access-control.spec.ts`
 - `tests/support/data-loader.ts`
 - `tests/support/fr12-api.ts`
- Rejected artifact:
 - `ai-audit/artifacts/0021-rejected-fr12-increment2/fr12-access-control.json`
 - `ai-audit/artifacts/0021-rejected-fr12-increment2/fr12-access-control.spec.ts`
 - `ai-audit/artifacts/0021-rejected-fr12-increment2/data-loader.ts`
 - `ai-audit/artifacts/0021-rejected-fr12-increment2/fr12-api.ts`
- Read-only SUT references needed to understand whether an API-only reversible order fixture is possible:
 - `backend/server.js`
 - `backend/database.js`

Correct only the rejected FR12-TC05 through FR12-TC08 proposal. Preserve all accepted TC01-TC04 code byte-for-byte relative to HEAD.

The rejection reason is specific:

TC05 and TC06 currently probe `PUT /api/admin/orders/:id/status` with a non-existing safe ID and treat any non-success response plus unchanged snapshots as authentication or authorization denial. If the access-control boundary is bypassed, the business handler can return `404 Order not found`; the current oracle can then pass falsely.

Required correction:

1. Do not treat a generic non-success response from the non-existing order-status target as proof of authentication or authorization denial.
2. First determine from the read-only SUT/API sources whether a controlled existing order can be established and fully restored using supported HTTP APIs only.
3. If an API-only reversible controlled order fixture is possible:
 - Create it through supported APIs, never by editing SQLite or injecting internal state.
 - Prove a strict positive-integer controlled order ID and its initial state.
 - Choose a business-valid transition whose payload differs from the initial state.
 - Execute each TC05/TC06 order-status target exactly once.
 - Assert that invalid-token and ordinary-user requests do not change the controlled order.
 - Record the original target response/state evidence before cleanup.
 - Restore all controlled state through authenticated Admin APIs in nested `finally`.
 - Verify the final observable state equals its pre-test baseline.
4. If no fully reversible API-only controlled order fixture exists:
 - Keep the safe order-status request one-shot and non-mutating.
 - Add explicit external JSON evidence-boundary data explaining that a non-existing target cannot distinguish access denial from business-handler `not found`.
 - Do not count generic non-success as a verified denied route.

- Produce an explicit guarded `EVIDENCE MISSING` assertion for that route so the test cannot pass falsely.
- Keep the response available for later human triage without asserting unsupported exact status or wording.
- Do not use skip, fixme, retry, arbitrary waits, mocks, storage injection, direct database access, or fabricated evidence.

5. Ensure aggregate route counters cannot report all seven routes as verified denials when the order-status authorization boundary was not actually proven.

6. Preserve the accepted design elsewhere:

- TC05: exactly two invalid-token classes and fourteen one-shot target probes across seven Admin routes per class.
- TC06: exactly seven one-shot ordinary-user target probes.
- TC07: exactly two GET-only Admin read targets.
- TC08: exactly nine POST/PUT/DELETE × missing/user/admin target rows.
- Original post-target defect evidence must be recorded before cleanup.
- Exact-marker discovery, strict positive-integer IDs, authenticated API cleanup, nested `finally`, and final baseline verification remain.
- No raw token or signing material may be logged.
- External test-case data remains in JSON; do not introduce inline test-case arrays or objects in the spec.

Permitted automation files:

- `tests/data/fr12-access-control.json`
- `tests/features/fr12-access-control.spec.ts`
- `tests/support/data-loader.ts`
- `tests/support/fr12-api.ts`

Do not modify:

- `tests/pages/admin-access.page.ts`
- FR01 or FR07 automation
- SUT source
- database files
- README, API specification, traceability matrix, or other documentation
- Playwright/TypeScript configuration
- package files or lockfiles
- evidence
- existing audit files, prompts, outputs, entries, or rejected artifacts

Validation requirements:

- JSON parses and IDs remain exactly sequential and unique FR12-TC01 through FR12-TC08.
- Exactly eight FR12 test declarations overall.
- TC01-TC04 remain byte-identical to HEAD.
- TC05-TC08 Playwright discovery lists exactly four tests for Chromium.
- Run `npm run typecheck`.
- Run Playwright only with `--list`; do not execute browser or API tests.
- Check one-shot target counts, evidence-before-cleanup ordering, guarded observations, forbidden patterns, trailing whitespace, and `git diff --check`.
- Confirm prohibited files are unchanged.

In the final response, state:

- the exact files changed;
- whether an API-only reversible order fixture was possible;
- how the false-pass order-status oracle was eliminated;
- TC05 through TC08 structure and action counts;
- cleanup and evidence boundaries;

- static validation performed;
- that no browser/API test ran and no prohibited file changed.

Exact AI output

```text

Implemented the corrected FR12-TC05-TC08 proposal.

Changed files:

- [fr12-access-control.json](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/data/fr12-access-control.json)
- [fr12-access-control.spec.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/features/fr12-access-control.spec.ts)
- [data-loader.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/data-loader.ts)
- [fr12-api.ts](/Users/phamngocgiabao/HW04/eshop-sut-hw04-23127027/tests/support/fr12-api.ts)

An API-only reversible order fixture was not possible. `/api/checkout` creates a persistent pending order, but no supported HTTP endpoint deletes orders or reverses a status transition to its baseline state.

The false-pass oracle was eliminated by:

- Binding the three non-existing order-status probes to external JSON `EVIDENCE MISSING` definitions.
- Proving their safe IDs are strict positive integers and absent from the authenticated baseline.
- Retaining one-shot response diagnostics and post-target order snapshots.
- Never treating generic non-success, status, or wording as access-denial proof.
- Deliberately raising a guarded `EVIDENCE MISSING` assertion.
- Limiting verified-denial counters to six routes per token class, never seven.

Structure:

- TC05: 2 invalid-token classes × 7 routes = 14 one-shot probes; 12 potentially verified denials and 2 guarded order-status observations.
- TC06: 7 one-shot ordinary-user probes; 6 potentially verified denials and 1 guarded order-status observation.
- TC07: exactly 2 GET-only Admin reads.
- TC08: exactly 9 rows-POST/PUT/DELETE × missing/user/admin.

Original response/state evidence is recorded before exact-marker authenticated cleanup. Cleanup remains nested with strict positive-integer IDs and final collection-to-baseline verification.

Validation passed:

- JSON parsing and sequential unique FR12-TC01-TC08 IDs.
- Exactly eight test declarations.
- TC01-TC04 byte-identical to HEAD.
- Chromium `--list`: exactly four TC05-TC08 tests.
- `npm run typecheck`.
- Action-count, evidence-ordering, guarded-oracle, forbidden-pattern, logging, trailing-whitespace, and `git diff --check` checks.

No browser or API test ran. No prohibited file was changed; pre-existing `ai-audit` worktree entries were left untouched.

## Human review/action

Chấp nhận output 0022. Human action: Em đã review toàn bộ correction diff FR-12 increment 2 và xác nhận không thể tạo rồi khôi phục hoàn toàn controlled order fixture chỉ qua supported HTTP APIs vì checkout tạo persistent order nhưng không có order-delete endpoint hoặc reverse transition về baseline. TC05 và TC06 vì vậy gắn explicit external EVIDENCE MISSING boundary cho đúng hai invalid-token và một ordinary-user order-status probes. Các safe order IDs được hard-check là strict positive integers và vắng mặt trong authenticated baseline; mỗi target vẫn chạy đúng một lần và giữ original response/post-target snapshot evidence, nhưng generic non-success, status hoặc wording không còn được xem là access denial. Guarded EVIDENCE MISSING assertions ngăn test pass sai; counters chỉ cho phép TC05 có 12 potentially verified denials, sáu mỗi token class, và TC06 có sáu potentially verified denials. TC05 vẫn có đủ 14 probes, TC06 đủ bảy routes, TC07 đúng hai GET-only reads và TC08 đúng chín POST/PUT/DELETE x missing/user/admin rows. Evidence-before-cleanup, exact-marker authenticated cleanup, strict positive-integer IDs, nested finally và final baseline verification được giữ nguyên; fr12-api.ts byte-identical với rejected artifact 0021. TC01-TC04 byte-identical với HEAD, FR01/FR07 và Admin Page Object không thay đổi. External JSON có IDs tuần tự FR12-TC01 đến FR12-TC08, spec có đúng tám declarations, typecheck thành công và Playwright chỉ liệt kê đúng TC05-TC08 trên Chromium. Forbidden-pattern, whitespace và tracked diff checks đều sạch. Chưa chạy Playwright browser/API test và không sửa SUT, database, evidence, configuration, documentation hoặc package files.

# Interaction 0023

## AI tool

OpenAI Codex

## ISO date/time

2026-08-21T15:12:00+07:00

## Exact prompt

```text

Mình vừa đặt file tên HW04_Testing.pdf vào repo này bạn hãy đọc tình trạng hiện tại và so với các yêu cầu của file xem còn thiếu phần nào thì làm tiếp cho xong giúp mình, các phần nào cần mình cần tự làm thì hãy để dành sau khi bạn xong hết rồi nói tôi để làm từng bước để hoàn thành toàn bộ file bài tập này

Exact AI output

See `ai-audit/outputs/0023-complete-hw04-submission.md`.

Human review/action

Chấp nhận output 0023. Em đã đọc kết quả kiểm tra từng bước và đồng ý với phần review code, cấu hình, tài liệu và execution evidence. Bộ test có đủ 36 logical cases, gồm FR12-TC01 đến FR12-TC12; các mutation có cleanup, TC05/TC06 giữ đúng phân loại EVIDENCE MISSING và không có skip, retry hay fixed sleep để né lỗi. Signing material đã được chuyển khỏi JSON sang biến môi trường; giá trị cũ trong rejected artifact cũng được ghi rõ là redacted trước khi public. Report local được tách tại `playwright-report/local`, còn chịn report final được giữ riêng tại `playwright-report/final`; evidence hiện có tổng cộng 108 executions với 30 pass và 78 fail. Nội dung final report, bug report, AI critique và workflow skill phù hợp với evidence đã tổng hợp. Em đã tự tạo 16 GitHub Issues kèm ảnh evidence và đã yêu cầu Codex thực hiện các thao tác kỹ thuật commit, cập nhật Git log, push branch và cài workflow skill. Video vẫn để PENDING để em tự quay bằng giọng nói và bằng chứng danh tính của mình; em cũng sẽ tự đóng gói và nộp bài. Em chấp nhận

n trạng thái hiện tại của output sau các chỉnh sửa trên.